

OptiPrime: Development of a Three-Domain Autonomous System for the 2026 Maritime RobotX Challenge

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ABSTRACT

Team OptiPrime of the Singapore Institute of Technology developed an Uncrewed Surface Vehicle (USV) "Impavida", an Uncrewed Underwater Vehicle (UUV) "Riptide" and an Uncrewed Aerial Vehicle (UAV) "Buzzlight" for the 2026 Maritime RobotX Challenge. The strategy targets the Core tier of all four mission tasks with the surface and underwater vehicles and engages the UAV only at the Advanced tier, so that Core-tier progress never depends on the aerial link. Impavida is a BlueBoat hull with two added lateral thrusters and custom ArduPilot firmware for the docking sway movement, Riptide an untethered positively buoyant eight-thruster vehicle with a hydrophone array, an acoustic modem and a magnetic repair probe, and Buzzlight a quadrotor carrying a pre-loaded resource tin with a servo release. Impavida is the hub of the intervehicle network, and on every vehicle perception runs on its own computer and returns coordinates but not imagery. USV perception uses two stereo cameras of different focal lengths, a long-range unit for buoy detection and a short-range unit for docking and water cannon aiming. Path generation uses a Dubins planner with rotate-in-place as the normal case and an obstacle-aware planner as fallback, selected on a benchmark of six planners. Verification of performance follows a ladder from a kinematic simulator through Gazebo, bench unit tests and water trials. Impavida completed flotation, manual-control and autonomous waypoint navigation trials in August-September 2026 and Buzzlight completed flight and stability trials in the same period.

I. COMPETITION STRATEGY

A. Strategic vision

OptiPrime is a newly formed team and enters RobotX 2026 with a system of systems covering three domains: a commercial surface hull, an underwater vehicle built from watertight enclosures, and a quadrotor built on a carbon frame. The RobotX rules call for a USV and at least one further vehicle in another domain [1]. The team chose to field all three so that every mission task can be attempted with the vehicle best suited to it. Impavida carries surface transit, docking, logistics and surface incident response. Riptide carries submerged infrastructure survey and repair. Buzzlight carries aerial search and localization, cross-domain cueing, resource delivery and air-domain incident response.

Because every system was built in one campaign, the design leans on a commercial hull and on module boundaries that let a new team work in parallel: a Global Mission Supervisor assigns tasks, each vehicle runs its own hierarchical state machine, and module contracts were fixed and versioned before work was distributed.

The objective for 2026 is reliable completion of the Core tier of Safe Passage, Infrastructure Survey and Repair, Coordinated Logistics and Dynamic Incident Response. Advanced and Disruptive tiers remain in the team's focus and are attempted where the enabling capability has been demonstrated (table 1). The UAV was deliberately confined to the Advanced and Disruptive tiers, so that Core-tier effort is never blocked on aerial autonomy or on the aerial link.

TABLE I
TASK AND TIER SELECTION

Mission task	Systems at Core	Core target	Extension
1. Safe Passage	USV	Autonomous transit entry to exit with correct passing sides	Advanced: UAV overhead route cueing. Disruptive: UAV on station, within battery endurance
2. Infrastructure Survey and Repair	USV, UUV	UUV survey and ordered status report; magnetic-probe repair for additional credit	Advanced: resource request from the UUV, pre-loaded tin delivered by the UAV
3. Coordinated Logistics	USV	Dock in the GREEN bay, water delivery to the RED window	Advanced: resource request from the USV, pre-loaded tin delivered by the UAV
4. Dynamic Incident Response	USV, UAV	Acknowledge, re-task, loiter, report, resume	Advanced: UAV static keep-out area. Disruptive: UAV moving object at 10 m. All tiers targeted

At Core tier of Task 2 the entry buoy is localized by Impavida rather than the UAV and autonomous tin retrieval (the Disruptive extension of Tasks 2 and 3) is not targeted.

B. Complexity versus reliability

Available time before the competition were the main constraint. The following decisions traded capability for reliability, and each is carried into Section II.

1. **Commercial USV hull, one structural change.** The BlueBoat was chosen over a custom build so that effort went into integration and testing, with the only change is two lateral thrusters, used on the docking search.
2. **Simplest viable compute.** Controller, supervisor and communications start on one board and split only if resources prove inadequate. vision was isolated from the start.
3. **Pre-loaded tin, mechanical release.** The tins are aluminum, so the passive magnet considered for Buzzlight fails. A pre-loaded tin pushed by one servo removes the retrieval stage entirely.
4. **Fielded sensors only.** The Livox LiDAR is installed and its pipeline works on live data, but it is out of the mission chain because its on-water test could not be scheduled in time.
5. **Tiers tied to evidence.** A tier is attempted only if its critical path has been exercised in trials or simulation. For example, Task 1 Disruptive tier depends on the UAV's endurance covering a full transit without search pauses.

C. Approach to the course

Every task runs in one frame: the OCS declares the run, publishes heartbeats for all vehicles at 2 Hz and relays every task report to RoboCommand [1]. Full task sequences are in Appendix C.

Task 1, Safe Passage. Impavida circles the entry clockwise, transits with RED to starboard and GREEN to port, and circles the exit counterclockwise, on stereo vision alone: detections become landmark tracks, the path is planned gate by gate because the exit is unknown, and bearing-only detections are search leads. LiDAR is excluded because a buoy at range returns too few points on a moving hull. Running Core tier without the UAV was risky, since from water level buoys might line up behind one another, but the team kept the decision and performed the inland test, with two buoys one behind the other. At Advanced tier Buzzlight detects the overhead beacons on its own companion computer and sends only coordinates, so a dropped link delays the route rather than losing it.

Task 2, Infrastructure Survey and Repair. Impavida localizes the active buoy and hands its coordinates to Riptide over the acoustic modem. Riptide acknowledges and transits on the surface with GNSS, submerges vertically short of the buoy, homes on the pinger with the hydrophone array, follows the pipeline in the image plane logging each ring GREEN or RED, returns to the RED ring, holds the

magnetic probe until it turns GREEN, decodes, and surfaces. Segment order comes from pinger intensity sampled as each ring passes a fixed image window, with position ordering as fallback. In the mission simulator, which supplies ring colors as metadata, this sequence completed 100 of 100 runs across three scene types and twenty initial headings (Appendix C). At Advanced tier Buzzlight, already airborne with a tin loaded, delivers on Riptide's decoded request.

Task 3, Coordinated Logistics. Impavida rotates in place until the dock is recognized, ranges the three bays, fits a line through them on the map and offsets it, sways along that line under CUSTOM_AUTO recording every bay before committing, returns to the GREEN bay and approaches with thrust cut at a standoff. The close range camera then localizes the RED window for the cannon. The Advanced extension mirrors Task 2.

Task 4, Dynamic Incident Response. An assistance request is an overlay on the running task: its state is preserved, the incident sequence runs, and the task resumes where it stopped. A keep-out area enters the map as a planner constraint and a moving object is a moving obstacle for the same planner. All three tiers are targeted because each extends the same executive rather than adding a subsystem.

II. DESIGN STRATEGY

A. System of systems

Impavida (BlueBoat USV), Riptide (custom untethered UUV), Buzzlight (quadrotor UAV) and an Operator Control Station (OCS) form the system. Impavida is the hub of the team network. Buzzlight and Riptide report to the Global Mission Supervisor on the boat, and the boat's OCS report path is the route to RoboCommand. The OCS declares the run, publishes heartbeats for all vehicles at 2 Hz, and relays report of every task. (Figure 1 – models of the vehicles).

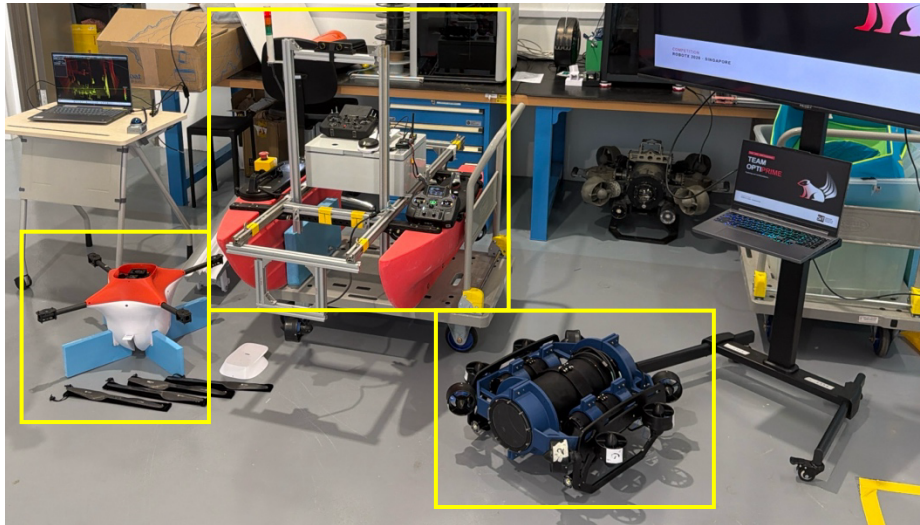


Figure 1. Early assembly of Impavida, Riptide and Buzzlight at SIT Student Innovation Center.

B. Impavida: platform and sensors

Impavida is a Blue Robotics BlueBoat [2] with an aluminum superstructure, 1.42 m by 1.00 m by 1.00 m, inside the 2 m by 1 m by 1 m and 60 kg limits as required by the rules [1]. The hull arrives waterproof and tuned for differential thrust, so integration effort went into sensing and autonomy. Two T200 lateral thrusters were added at bow and stern, symmetric about the center of gravity so that a pure sway command produces negligible yaw moment. Sensors are a 4 mm ZED 2i with a polarizing filter for range, a ZED Mini for close targets, a Livox Mid-360 LiDAR (outside the mission main chain), and a Holybro H-RTK ZED-F9P rover with an M10 V2 secondary GNSS (Figure 2). Power is a four-cell

lithium bus with a Kilovac contactor in the kill path. System architecture for USV is illustrated on figure 2.

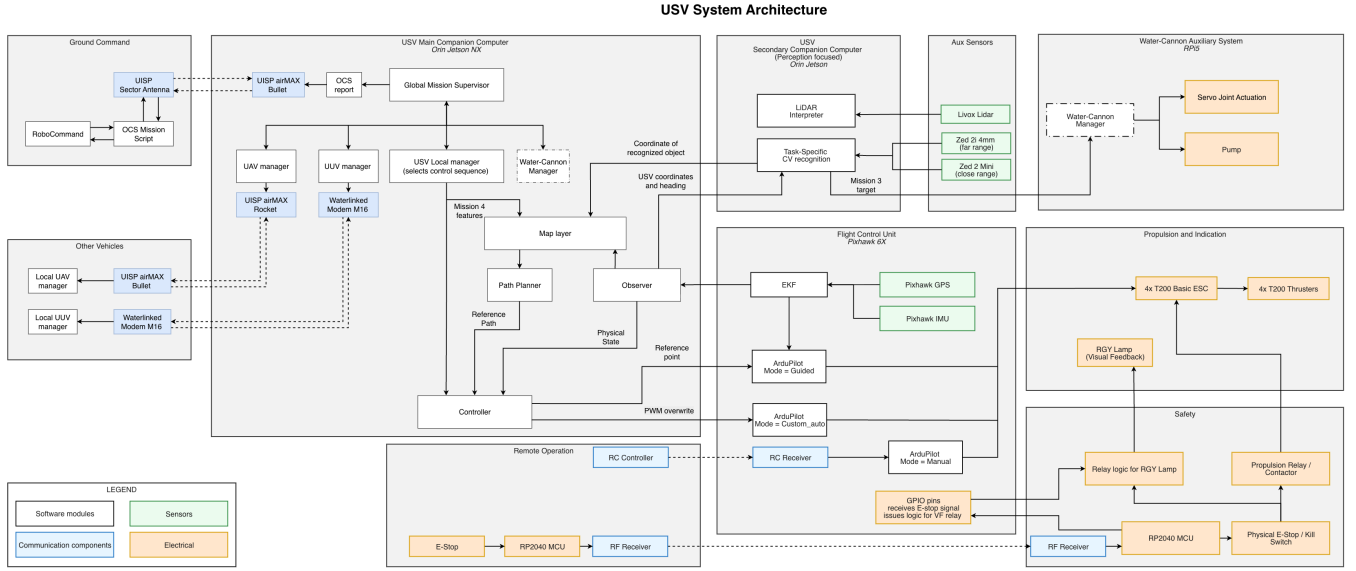


Figure 2. Impavida system architecture.

C. Compute partitioning

Vision was isolated on evidence: a continuous vision workload measured on a Jetson AGX Orin (used for CV task on another project) consumed enough compute so it was decided that planning and control must not share a board that can saturate under it. Layers run at their own rates so that the supervisory loop never constrains the control loop. Riptide uses the similar pattern of compute partitioning.

TABLE II
IMPAVIDA COMPUTE PARTITION

Element	Role	Rate
Pixhawk 6X Pro	Low-level control, thruster output, EKF pose source	Autopilot native
Jetson Orin NX 16 GB	Supervisor, vehicle managers, map layer, planner, observer, controller, reporting	Supervisor at or below 1 Hz, controller above 10 Hz
Jetson Orin Nano, perception	Task-specific detector, LiDAR interpreter	Frame skipping rate
Raspberry Pi 5	Water-cannon manager, servo and pump	Not rate critical

D. Custom ArduPilot firmware

Default ArduRover [3] has no frame for two aft plus two lateral thrusters and cannot command sway, so the team forked ArduRover 4.7 for Impavida, while Buzzlight and Riptide run stock ArduCopter and ArduSub. The OmniPlus frame type was modified to keep the stock skid-steer allocation for the aft pair, preserving the transit tuning established on the water, and to add the lateral pair with pure-sway factors. Additional custom mode was added, CUSTOM_AUTO, which accepts per-motor thrust from the companion computer, and Remote Controller (RC) thrust input ignored. RC switch toggles it against MANUAL, so the operator regains direct control in one action and no state mixes RC and autonomous commands. Stock GUIDED serves transit stages.

E. Perception

Interface. The vision computer returns the coordinate and color of detected objects and never imagery, and own pose is pushed to it to stamp each frame, so a detection processed after capture is

placed at the pose the vehicle had at exposure. This keeps perception latency out of the control loop and the map independent of the perception implementation.

Two cameras, chosen by sweep. The original 2.1 mm ZED 2i was insufficient at the 30 m range which Task 1 needs. In the kinematic simulator a sweep of 1,800 course layouts showed the 4 mm unit (72 deg field of view, about 25 m usable depth) handling every turn to 16 m gate spacing and most to 24 m, while the 2.1 mm unit was spacing-limited beyond 16 m, and a 540-run sweep showed nothing working below 20 m of depth range on the hardest course at any field of view. The limitation of Task 1 is a camera range, which justified a second camera rather than an extended controller parameter search: stereo depth is $Z = f_x B / d$, so a fixed disparity error grows as Z^2 and a longer focal length is the only lever on it (Appendix D). The ZED Mini serves the close range tasks of Mission 3. Range tests against tape-measured distance cones from 5 to 35 m, camera depth agreed with true distance to a mean absolute deviation of 0.41 m (Figure 3), with detections above 30 m stable in value but flickering, with one caveat until the over-water sweep runs: water is low-texture, specular and moving.



Figure 3. Camera range against true distance.

The cameras classify on color and shape and provide stereo range, which a point cloud cannot, while a small floating target returns too few LiDAR points at range under platform motion, so the LiDAR's one strength, plane extraction on a large structure, is held in reserve as a backup plan.

Detectors. All three vehicles run team-trained Ultralytics YOLO networks [4]. YOLO11n is used as the development model and YOLO26-L is used as the competition model. Impavida's development model is a YOLO11n fine-tune on cones composited onto pool, indoor and noise backgrounds with red and green distractors so the network cannot rely on color alone. It scores 98.9 % precision and 100 % recall on held-out frames after an indoor-only pilot had transferred at 49.1 %. Riptide's YOLO11n was trained on the 2,200-image UW-Light-Detection Benchmark [5], [6] and reaches 82.4 % mAP at 0.5 to 0.95 on the held-out split from a 5.5 MB checkpoint. Buzzlight runs a YOLO11-L with color detection. Depth distance is read from stereo depth inside a central sub-region of each box after a class HSV mask and outlier rejection, ring range underwater from projected area, and flash state over at least one full cycle (Appendix D).

F. Mission management, map layer and path planning

A Global Mission Supervisor on Impavida assigns tasks to Buzzlight, Impavida and Riptide through vehicle specific UAV manager, a UUV manager and a USV local manager, and tracks vehicle, mission and disruption states. Each vehicle runs a local manager and hierarchical state machine that selects and executes the control sequence and reports started, in progress or accomplished. The executive is layered:

safety and authority at the bottom, run control, the task executive with its incident overlay, and world-model constraints supplied to the planner as inputs. Reported vehicle state derives from the supervisor's authority state after receiving a confirmation from vehicle specific manager.

Map layer. A distinct module holds the absolute map with detected objects as landmark tracks and used as a basis for generating a reference path, a geometric path with no timing. Tracks carry a covariance, are associated by a Mahalanobis gate and fused by inverse variance so the estimate follows the EKF's own drift [7]. Only classified tracks with position uncertainty below 1.5 m are planned on, and bearing-only detections are held as ray tracks that never update a position until a depth detection lands on their bearing. Gates are red-green pairs with pre, cross and post waypoints so the crossing is perpendicular, and the exit is appended only once no unresolved leads remain.

Planner. USV planning has a large literature and few field-validated planners [8]. Six candidates were benchmarked in simulation against the same pose, target and known obstacle map, each planning only to the current objective with unknown exit, over 30 seeds per condition with exact geometry.

TABLE III
PLANNER BENCHMARK, COLLISION-FREE PATH RATE

Condition	Dubins, rotate-go [9]	A* [10]	Hybrid A* [11]	D* Lite [12]	RRT* [13]	Visibility + A* [14]
Baseline	100 %	100 %	100 %	100 %	100 %	100 %
Sharp turn	73.3 %	96.7 %	97.8 %	96.7 %	97.2 %	97.2 %
Full stress	44.4 %	97.2 %	98.9 %	96.7 %	98.3 %	98.3 %

Median solve times were 0.04 ms for Dubins against 5 ms for A* and D* Lite and 24 to 36 ms for the sampling and graph planners. Although Dubins produced the shortest median solve time, its collision-free rate degraded substantially in sharp-turn and full-stress cases. It is therefore used only as the nominal planner when the resulting path passes the clearance check. If the path fails that check, the system invokes A or D Lite as the obstacle-aware fallback. This preserves deterministic low-cost planning in uncomplicated geometry while retaining obstacle-aware planning for constrained cases. When no gate is observable the search investigates bearing-only leads, then a heading-aligned outward spiral, then frontier exploration, since a stationary scan cannot solve a range problem. At hard course layout difficulty, the spiral completed 86.7 % of runs against 66.7 % for scanning. Riptide's pipeline following runs in the image plane, from the mean and principal axis of the detected ring centers (Appendix E).

G. Guidance and control

The control algorithm runs on the companion and the low-level controller on the flight controller, linked by a Motion API over MAVLink [15] (Figure 4). The observer module wraps the FCU's EKF3 solution rather than running a second estimator. Its main function is a pose-history buffer queryable by timestamp, for example by CV module to estimate the global coordinate of recognized object. Surface and air vehicles carry an RTK-capable primary GNSS with a base station in the ground kit. Control architecture is illustrated on figure 3.

Guidance is line of sight [16], [17] with adaptive look-ahead and integral action for sideslip on Dubins paths [18]: The implemented guidance architecture uses LOS path following on the companion computer, with the Pixhawk retaining the low-level control loops. Bench and simulation tests verified the guidance chain. During the August water trial, however, the companion-side controller did not maintain the commanded track, and the Proof-of-Readiness run therefore used autopilot waypoint navigation. Subsequent work focuses on resolving the interaction between companion-generated targets and the GUIDED-mode heading response before repeating the water validation. Targets streamed to GUIDED carry velocity as well as position, since a bare position target implies arrive-and-stop and a

stream of them risks producing a sawtooth type PWM behavior. Table IV describes implemented modes and their features per mission state.

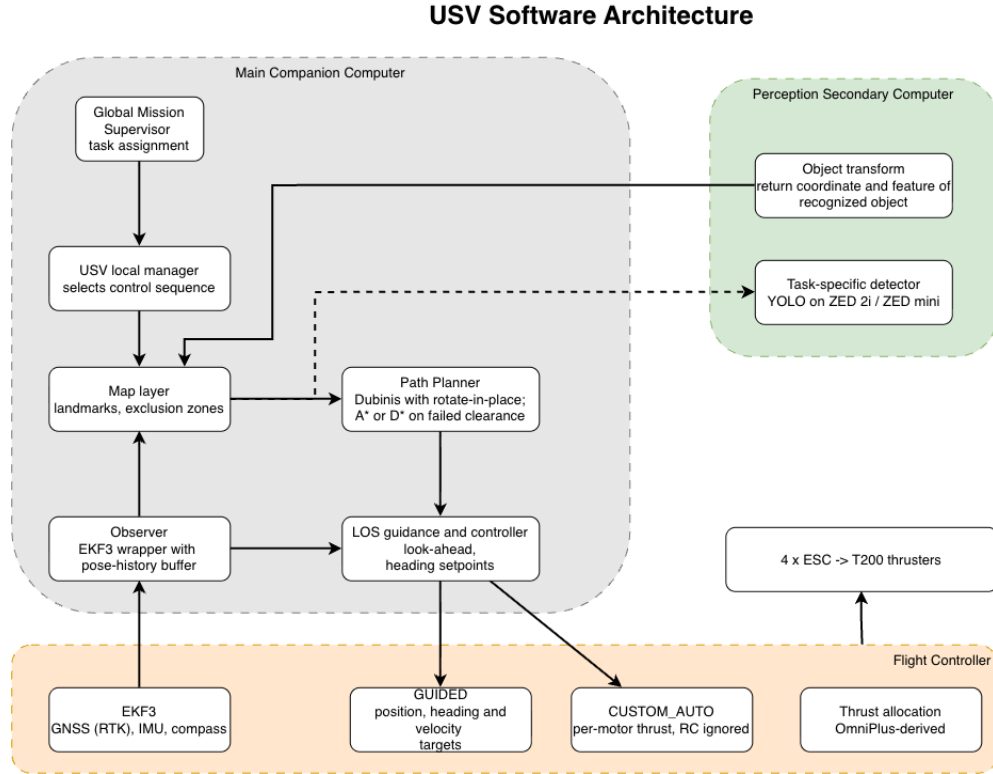


Figure 4. USV control architecture: supervisor, local manager, map layer, planner, LOS guidance, observer, perception computer and flight controller modes.

TABLE IV
USV FLIGHT CONTROLLER INTERFACE BY MISSION STATE

Mission state	FCU mode	Command
Transit (Task 1 gates, Task 2 entry, Task 4 diversion)	GUIDED	Position and velocity targets
Approach on bearing without range	GUIDED	Heading-only servoing
Docking sway (Task 3)	CUSTOM_AUTO	Per-motor thrust from companion-side allocation

GUIDED was preferred over AUTO because it keeps the tuned inner loop while the companion owns the guidance decision, and because the rules require the autonomy to run on the onboard companion rather than as a pre-loaded waypoint mission [1].

H. Riptide

Riptide is a custom untethered vehicle on eight T200 thrusters, trimmed positively buoyant and holding depth on downward thrust, so any loss of propulsion returns it to the surface with no software involved. Two batteries feed two independent buses, propulsion through a kill circuit with a Kilovac contactor and electronics through a PM02D module with a regulated 9 V rail for the hydrophone preamplifiers, so a kill isolates propulsion and nothing else, a killed vehicle keeps reporting through its ascent, and thruster transients stay off the sensor rails.

Riptide mirrors Impavida's partition: a Jetson Orin NX for perception, with the low-light camera, four Aquarian Audio S1n hydrophones through PA4 preamplifiers into an MCC USB-1608FS-Plus at 100 kS/s and an A50 DVL, a Pixhawk 6X Pro for thruster and depth control through a Matek CAN-L4-PWM node with Bar30 pressure, leak sensor and H-RTK GNSS, and a Raspberry Pi 5 as mission computer

with M16 acoustic modem. The three computers and the DVL share a GigaBlox Ethernet switch, and MAVLink between mission computer and flight controller runs over it. Underwater, heading is the weak point: the magnetometer sits inside a metal tube and the gyroscope drifts on slow turns, so vision is the relative reference submerged and pressure depth the absolute one, which is why the pipeline is followed based the camera feed. The DVL can be fused with the IMU to bound position drift but not heading. The hydrophone array feeds the observer with a fix relative to the entry/start pinger. The transducer is mounted aft so the vehicle presents its stern to Impavida, and probe contact is confirmed by the camera when the ring turns GREEN (Appendix H).

I. Buzzlight

Buzzlight is a quadrotor on a ZD680 folding carbon frame with four T-Motor U5 KV400 motors and AIR 40A ESCs on a 6S battery, the same Pixhawk Jetson baseboard as Impavida, dual GNSS, the same low-light USB camera as Riptide, and a payload servo driven from the onboard computer. It runs stock ArduCopter and all of its perception on board. The delivery mechanism is a cradle holding a pre-loaded tin with a servo that pushes it clear, one moving part and no sensing, so its failure modes reduce to release timing and drop accuracy over a 60 to 70 cm circle, set by ground drop tests [1]. The design risk is the communication path to the other vehicles, not the flight, and endurance sets the Task 1 Disruptive decision.

J. Communication architecture

On the ground a router is the hub, with the OCS, one ground station per vehicle and a 5 GHz sector radio on Ethernet. Every vehicle has up to four radio paths: a primary IP link at 5.8 GHz carrying MQTT [19] with Protocol Buffers payloads [20], a 915 MHz MAVLink telemetry link to its own ground station, a safety link with an RP2040 at each end, and manual RC. Impavida holds the primary link to the ground and a second 5.8 GHz radio that is Buzzlight's link, and Riptide reaches the boat by acoustic modem submerged. Message schemas follow the RoboCommand interface control document [21], so one message set serves internal coordination and judge-facing reporting, and every module obeys one set of conventions for frames, units, time stamps, identity, uncertainty and envelope, fixed before work was distributed and versioned from the first schema (Appendix F).

The submerged link was sized with the passive sonar equation [22] rather than the vendor's rated range. At 58.5 kHz, with absorption about 17.5 dB/km in seawater [23], the loss relative to a 50 m reference is +7.7 dB at 100 m and +15.4 dB at 200 m, the largest Task 2 separation. That is small against the modem's source level, so geometric spreading and absorption are not expected to be the dominant limitation at course distances; venue noise and shallow-water multipath remain to be validated experimentally. However, the venue noise, of order 70 to 80 dB re 1 μ Pa in the receive band before harbor traffic [24], and shallow-water multipath do, and multipath is also the one condition that can defeat the intensity based ordering in Task 2. The design therefore keeps the acoustic exchange to one acknowledgement and treats a venue noise measurement as the one input the model cannot supply (Appendix F).

K. Safety and operator authority

Propulsion kill on Impavida is in hardware: an RP2040 drives a Kilovac contactor on the propulsion rail, commanded by the onboard pushbutton and by a dedicated 915 MHz radio pair with an RP2040 at each end, independent of telemetry and IP links and acting within 2 s of link loss; the flight controller is not in the kill path. The status lamp follows the rules, with RED asserted from the contactor in hardware so a software fault cannot show a safe color on a killed vehicle. Operator takeover is one RC switch to stock MANUAL. Riptide carries its own kill contactor and subsea indicator, and its positive buoyancy makes every kill a controlled ascent with the electronics bus live.

III. TESTING STRATEGY

Testing runs in three streams: bench tests for interface handshakes and component operation, simulation at two levels of fidelity, and water trials per vehicle per mission stage, feeding individual Core trials and an integrated rehearsal before any higher-tier attempt.

A. Verification ladder

Every capability climbs the same ladder and no stage is skipped because an earlier one passed cleanly.

TABLE V
VERIFICATION LADDER

Stage	Retires	Notes
Kinematic simulator	Perception-to-navigation logic	No hull dynamics, upper bound on success
SITL and Gazebo	Mode logic, state machines, allocation, sequencing	Flight controller in the loop
Bench	Wiring, links, heartbeats, per-component function	One layer at a time with a pass or fail gate
Water	Hydrodynamic coupling, actuator authority, underwater sensing	Pool, then venue water

B. Bench and water

Each subsystem was bench tested one layer at a time, with Lua probe scripts on the FCU and pymavlink STATUSTEXT listeners on the companion as diagnostics. Water trials are ordered by the uncertainty each retires: acoustic received level and packet loss against range, hydrophone bearing on a printed jig, LED-ring mockups on the bottom, a closed-loop approach on the tether, Impavida's sway sweep past a PVC dock mockup, a kill at depth to record Riptide's unaided ascent, and Buzzlight's loiter endurance and drop accuracy on the ground. Riptide's first water trials were tied to its Proof of Readiness, due 12 October. Venue water in Marina Bay or Kallang served color calibration under real light, a pinger reflection test and an ambient noise recording (Appendix G).

C. Simulation

The Task 1 chain was developed in a purpose-built kinematic simulator that models the problem at the detection level: stereo range noise growing as $a + b r + c r^2$ from the datasheet bound, miss probability rising toward the edge of range, a bearing-only band beyond usable depth, flashing lights, and a drifting EKF frame in which everything the boat computes lives while only scoring uses truth. Hydrodynamics are not modeled, so its rates are upper bounds. Three sweeps, 780 tuning runs, 1,800 course layouts and 540 camera ranges, produced the camera choice decision and a tuning result: the gate-approach corridor half-width is the decisive parameter, collapsing below 0.4 m under 0.5 m of drift. Riptide's mission was developed in a second simulator with four hydrophone channels at the physical array spacing, which completed 100 of 100 missions across three scene types and twenty headings. Gazebo Harmonic [25], [26] with the flight controller in the loop holds Impavida and Buzzlight and validated the LOS controller before the first autonomous water trial (Appendix E).

D. Trials completed and what they changed

TABLE VI
TRIALS COMPLETED NEAR KALLANG WAVE, SINGAPORE

Date (2026)	Vehicle	Trial	Outcome
17 and 18 Aug	Buzzlight	Flight and stability	Completed
19 Aug	Impavida	Flotation and manual control	Completed
26 Aug	Impavida	Autonomous navigation	Completed, Proof of Readiness video recorded

Date (2026)	Vehicle	Trial	Outcome
22 Sep	Impavida perception chain, on land	Camera, detector, observer and map with cones	4,504 observations logged
23 and 24 Sep	Impavida perception chain, on land	4 mm camera, GPS status logged, two-cone separation, occlusion, range against measured distance (fig.4) from 5 to 35 m	Mean absolute range deviation 0.41 m over 11 readings

Eight findings from August-September period impacted on the design (Appendix G). The companion-side controller, which had passed on the bench, did not hold its track in water and the Proof of Readiness run used autopilot waypoints. The suspected cause is a conflict between the companion's heading setpoints and the autopilot's own heading response in GUIDED, and the fix is tested in SITL before the next water trial. The occlusion test found the physical perception boundary: separated cones resolve as two landmarks, near-total occlusion merges them, carried as an operating constraint.

E. Time allocation

Three rules protected testing time. The Proof of Readiness deadline of 31 August took priority over new capability, and simulation was suspended for that push with a named resume date. A pool session costs about five hours of transport and setup, so perception and mapping are tested on the grass outside the campus first. And a sensor that could not be tested on the water in time was left out of the mission chain rather than fielded untested.

IV. CONCLUSIONS

OptiPrime developed a three-domain system of systems for RobotX 2026 with the objective of completing all four Core tasks. The decisions that carry the most weight are compute partitioned by criticality and loop rate, a perception interface that returns coordinates rather than images, a second stereo camera chosen on a simulation sweep and confirmed against tape, a deterministic map feeding gate-by-gate Dubins planning and a network with the surface vehicle as hub. Impavida and Buzzlight completed their initial trials in August 2026, and eight findings from simulation, trials and land tests changed the design. The next stages are Riptide's water trials, Impavida's sway and docking trials, and an integrated Core rehearsal with RoboCommand reporting before any higher-tier attempt.

ACKNOWLEDGEMENTS

Team OptiPrime would like to express our sincere gratitude to Prof. Victor and Prof. Bernard for their invaluable guidance, mentorship, and continuous support throughout our preparation for the 2026 Maritime RobotX Challenge. Their expertise, encouragement, and constructive feedback have been instrumental in guiding our team through the design, development, and integration of our autonomous systems.

We would also like to extend our heartfelt appreciation to our sponsors and supporting organizations: Office of Naval Research (ONR), ALT TAB, iLaser, BeeX, Ministry of Defence (MINDEF), RoboNation, TGN Technology, Water Linked, and VectorNav. Their generous contributions, technical expertise, and support have provided us with valuable resources and opportunities to advance our project.

Finally, we would like to thank the Singapore Institute of Technology (SIT) for providing the facilities, resources, and opportunities that have enabled our team to apply our engineering knowledge to real-world maritime challenges.

The collective support of our professors, sponsors, and institution has been instrumental in bringing Team OptiPrime's vision to life. We are deeply grateful for their trust and commitment to nurturing the next generation of engineers and innovators.

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APPENDIX A: CORE MISSION TEST PLAN AND INTEGRATION

TABLE A1
REQUIREMENT ALLOCATION AND VERIFICATION STAGE

Task	Requirement	Capability	Allocated to	Verified at
1 Core	Classify lit buoys, pass with correct sides	Stereo detection and range, gate-by-gate planning, LOS tracking	ZED 2i 4 mm, map layer, planner, ArduPilot GUIDED	Simulator, Gazebo, water
1 Advanced	Overhead-only route	Overfly detection, route hand-off	Buzzlight camera and Orin NX, air link	Field
2 Core	Localize buoy and pinger, follow pipeline, report in order	Acoustic hand-off, hydrophone homing, ring tracing, intensity ordering	ZED 2i, M16 modem, Riptide acoustic array and camera	Simulator, Water
3 Core	Find GREEN bay, dock, spray window	Dock line on the map, sway sweep, cannon aiming	ZED Mini, lateral thrusters, cannon on Pi 5	Bench, Gazebo, water
2 and 3 Advanced	Decode flash, relay, deliver a tin	Pattern decoding, inter-vehicle tasking, servo release	Vision boards, MQTT, cradle	Bench, drop test, field
4 all tiers	Acknowledge, divert, resume, avoid zones	Incident overlay, zones as planner inputs	Supervisor, state machines, OCS	SITL, Gazebo, rehearsal
All	Reports via OCS, heartbeats, kill in 2 s, lamp	RoboCommand ICD, hardware kill, lamp	OCS, RP2040 and contactor	PoR build, bench

TABLE A2
CORE MISSION TEST PLAN

Task	Core completion requirement	Planned test and record
Safe Passage	Report the flashing-BLUE entry, steady-BLUE exit and detected buoy states. Circle the entry clockwise, transit with flashing RED to starboard and GREEN to port avoiding obstacles, circle the exit counterclockwise.	Randomized ten-buoy layouts including unlit obstacles and one-second flashes. Compare detections, reported states and recorded path with ground truth. Log side and circling errors, contacts, completion time and interventions.
Infrastructure Survey and Repair	Locate the active GREEN buoy and pinger, find and follow the pipeline, report GREEN and RED segment status in order from the active-buoy end. Surfacing to report is permitted. Repair is additional credit.	Representative pipeline with known light patterns. Compare ordered reports with ground truth. Retain underwater video, vehicle path, detection errors, report receipt and elapsed time.
Coordinated Logistics	Dock fully in the GREEN bay and report. Locate the activated RED window, spray until GREEN, detect the five-second completion indication, report. RED bays are never selected.	Bench test of the spray system, then a representative dock with varied bay and window selections. Record docking accuracy, window selection, water delivery, light transition, reports, time and interventions.
Dynamic Incident Response	Acknowledge the domain-specific assistance request, interrupt the current task, reach the requested position, loiter, report readiness, resume after RoboCommand clearance.	Inject assistance requests during mission execution for each participating domain. Record command and acknowledgement timing, arrival error, loiter behavior, readiness and clearance messages, and correct resumption.

Advanced and Disruptive extensions [1]. Higher-tier trials follow the corresponding Core sequence.

- **Safe Passage.** Advanced: Buzzlight overflies the entry area, reports the overhead-only markers through the supervisor and the OCS and shares the route before Impavida enters. Disruptive: Buzzlight stays on station, the route changes during transit, and the record shows message age, replanning delay, route compliance and completion.

- Infrastructure Survey and Repair. First the Core magnetic-probe repair: holding the probe at the damaged segment turns its light GREEN and the task system registers the repair. Advanced: decode the one-second on and off destination flash (30 s) and deliver the pre-loaded tin to the correct circle.
- Coordinated Logistics. Advanced: decode the destination flash (60 s), relay to Buzzlight and RoboCommand, deliver the pre-loaded tin, report the mission associated with the delivery.
- Dynamic Incident Response. Advanced: inject a static keep-out area, verify acknowledgement, avoidance until All Clear, and acknowledgement of All Clear. Disruptive: inject a moving-object report, verify replanning against current and projected motion with at least 10 m separation, record minimum clearance and resumption.

Integration and rehearsal. Integration trials verify message formats, coordinate frames, timestamps, acknowledgements, timeouts, safe-state behavior and retained logs after the individual vehicle checks. Full rehearsals then combine the autonomous-mode transition, selected Core tasks, RoboCommand reporting, aborts and recovery. Message tests begin against a simulator and are then verified against the official course interface when it is available. For every run the team retains the hardware and software configuration, test identifier, conditions, ground truth, logs and video, task reports, elapsed time, successes per attempt, interventions, anomalies and corrective action. A trial passes when its mission sequence completes without an unsafe event or operator takeover, and is repeated across changed layouts and conditions before release.

APPENDIX B: COMPONENT TABLES

TABLE B1

IMPAVIDA (USV) COMPONENTS

Component	Vendor	Model or type	Custom or purchased	Cost (SGD)
Hull form and platform	Blue Robotics	BlueBoat Base Package and BaseStation; two side thrusters and a 3030 aluminum superstructure added	Purchased, modified; sponsored by RoboNation and ONR	6,279.68
Waterproof connectors	Blue Robotics	WetLink penetrators M10 5.5 mm LC (5), bulkhead adapters M10 to M14 (8)	Purchased	175.48
Propulsion	Blue Robotics	T200 thruster (4): two aft stock, two side added	Purchased, modified layout	614.40
Power system	Blue Robotics	Lithium-ion battery 14.8 V 18 Ah (4), power switch	Purchased with the hull package	0
Motor controls	Blue Robotics	Basic ESC (4)	Purchased	194.56
CPU	Holybro, NVIDIA	Pixhawk Jetson Baseboard: Pixhawk 6X Pro and Jetson Orin NX 16 GB (share of bundle)	Purchased	375.00
Teleoperation	Holybro, Radiomaster, Blue Robotics	Zephyr ELRS Gemini X receivers (2) with Radiomaster GX12; SiK 915 MHz telemetry; Xbox controllers (2) for BlueBoat manual control	Purchased	250.85
Compass	Holybro	H-RTK ZED-F9P rover puck (magnetometer)	Purchased	806.00
IMU	Holybro	Pixhawk 6X Pro internal set: ADIS16470, IIM-42652, ICM-45686, triple redundant	Included in CPU	0
Cameras	Stereolabs	ZED 2i 4 mm with polarizing filter (far range); ZED Mini (close range)	Purchased	1,559.72
Algorithms	OptiPrime	LOS guidance; Dubins with rotate-in-place; A* and D* Lite fallback; gate pairing and chaining; spiral search	Custom	In-house
Vision	Ultralytics, Stereolabs	Team-trained YOLO26-L buoy detector; ZED SDK stereo depth	Custom on open source	In-house
Localization and mapping	ArduPilot, OptiPrime	EKF3 with RTK GNSS; observer with pose-history buffer; landmark map layer	Custom on open source	In-house
Autonomy	OptiPrime	Global Mission Supervisor; USV hierarchical state machine; custom ArduRover 4.7 (OmniPlus-derived frame, CUSTOM_AUTO)	Custom	In-house
Open source software	ArduPilot, Ultralytics, OpenCV, Open Robotics	ArduRover 4.7, pymavlink, Ultralytics YOLO, OpenCV, MQTT and Protocol Buffers, Gazebo Harmonic and SITL, Mission Planner	Open source	0
Other: LiDAR	Livox	Mid-360 with splitter and adapter; installed, not in the mission chain	Purchased	725.00
Other: network	TP-Link	TL-SG108 8-port switch	Purchased	34.00
Other: E-stop and signalling	AVD, Jutech, TE Kilovac, Blue Robotics	Emergency button; TN2-BF-1A pushbuttons; Kilovac 500 A contactor in the kill path; Subsea RGB LED indicators (2)	Purchased	158.88
Other: water cannon	Kaichin, DFRobot	R385 12 V diaphragm pumps (2), 7 mm pipe, DFRobot SER0066 servos, PETG nozzle (4 mm exit) and mount, LM2596 buck, 7805 regulators	Custom on purchased parts	21.20

Component	Vendor	Model or type	Custom or purchased	Cost (SGD)
Total				11,194.77

TABLE B2
RIPTIDE (UUV) COMPONENTS

Component	Vendor	Model or type	Custom or purchased	Cost (SGD)
Hull form and platform	Blue Robotics	Custom pressure hull from watertight enclosures: tubes (3), O-ring flanges 75 and 150 mm, end caps and domes, clamps, trays, rails	Custom on purchased enclosures	2,376.96
Waterproof connectors	Blue Robotics	WetLink penetrators M10: 6.5 mm LC (9), 4.5 mm LC (4), 6.5 mm HC (2), 7.5 mm LC (5), blanks (4); assembly block	Purchased	353.47
Propulsion	Blue Robotics	T200 thruster (8); the heave set holds depth against positive buoyancy	Purchased	2,457.60
Power system	SmartAP, Elements, TE Kilovac, Blue Robotics, SkyRC	SmartAP PDB (2); 6200 mAh 4S LiPo (4); Kilovac 500 A contactor; battery cable sets (2); XY-3606 5 V converter; Q100 charger	Purchased, custom integration	1,112.34
Motor controls	Blue Robotics, Matek	Basic ESC (8); Matek CAN-L4-PWM node (DroneCAN to PWM)	Purchased	415.83
CPU	Holybro, NVIDIA, Raspberry Pi	Pixhawk Jetson Baseboard: Pixhawk 6X Pro and Jetson Orin NX 16 GB; Raspberry Pi 5 16 GB mission computer	Purchased	4,460.29
Teleoperation	Water Linked, Blue Robotics	Modem M16 (2, both ends of the acoustic link); Fathom ROV tether for tethered trials	Purchased	3,759.74
Compass	Holybro	Pixhawk 6X Pro internal; H-RTK ZED-F9P when surfaced	Included in CPU and GNSS	0
IMU	Holybro	Pixhawk 6X Pro internal set: ADIS16470, IIM-42652, ICM-45686	Included in CPU	0
DVL	Water Linked	DVL A50 (300 m, 3 m cable, interface board)	Purchased	5,530.85
Cameras	Blue Robotics	Low-Light HD USB camera	Purchased	153.60
Hydrophones	Aquarian Audio, Measurement Computing	S1n hydrophones (4), PA4 preamplifiers (4), USB-1608FS-Plus DAQ at 100 kS/s	Purchased, custom array mount	2,634.81
Algorithms	OptiPrime	Pinger bearing from the four-element array; pipeline ring tracing; acknowledge-and-go acoustic hand-off	Custom	In-house
Vision	Ultralytics, OptiPrime	YOLO11n ring detector; blob-area ranging; RED-to-GREEN contact confirmation	Custom on open source	In-house
Localization and mapping	ArduPilot, Water Linked, OptiPrime	ArduSub EKF3 with Bar30 depth and DVL aiding submerged, F9P GNSS surfaced; pinger localization	Custom on open source	In-house
Autonomy	OptiPrime	UUV local manager (hierarchical state machine) on the Pi 5, tasked by the Global Mission Supervisor	Custom	In-house
Open source software	ArduPilot, Ultralytics, OpenCV	ArduSub, pymavlink, Ultralytics YOLO, OpenCV	Open source	0
Other: sensors	Blue Robotics	Bar30 depth sensor; SOS leak sensors (4) with probes	Purchased	350.72

Component	Vendor	Model or type	Custom or purchased	Cost (SGD)
Other: network	BotBlox	GigaBlox SFP 6-port Ethernet switch	Purchased	289.28
Other: lighting and indication	Blue Robotics	Lumen subsea lights (2 sets, sponsored); Subsea RGB LED; Power Sense Modules (2)	Purchased	609.92
Total				24,505.41

TABLE B3
BUZZLIGHT (UAV) COMPONENTS

Component	Vendor	Model or type	Custom or purchased	Cost (SGD)
Hull form and platform	ZD680	Carbon folding quadrotor frame (3)	Purchased	253.94
Propulsion	T-Motor	U5 KV400 motors (10); P15x5 propellers (8 pairs)	Purchased	1,975.04
Power system	Elements, Holybro	6200 mAh 6S LiPo (2); solid-state 12500 mAh 6S (5); PM02D (2), PM03D (2), PDB XT30 (5), 5 V BEC (6)	Purchased	1,690.79
Motor controls	T-Motor	AIR 40A ESC (10)	Purchased	460.80
CPU	Holybro, NVIDIA	Pixhawk Jetson Baseboard: Pixhawk 6X Pro and Jetson Orin NX 16 GB	Purchased	3,980.00
Teleoperation	FrSky, Holybro	FrSky X20S with TD10 receiver; SiK 915 MHz telemetry; antenna masts (2); PoE injectors (4)	Purchased	188.17
Compass	Holybro	H-RTK ZED-F9P rover and M10 V2 (dual GNSS with magnetometer)	Purchased	806.00
IMU	Holybro	Pixhawk 6X Pro internal set: ADIS16470, IIM-42652, ICM-45686	Included in CPU	0
Cameras	Blue Robotics	Low-Light HD USB camera, the same unit as on Riptide	Purchased	153.60
Hydrophones	Not fitted			
Algorithms	OptiPrime	Delivery sequence (task, pick, ascend, GUIDED transit, descend, servo release, return); camera-assisted return to land	Custom	In-house
Vision	Ultralytics, OptiPrime	YOLO11-L detector with color detection: overhead beacons, gates, landing pad, delivery circle, flash-pattern decoding	Custom on open source	In-house
Localization and mapping	ArduPilot, OptiPrime	ArduCopter EKF3 with dual GNSS; gate positions reported to the USV map layer	Custom on open source	In-house
Autonomy	OptiPrime	UAV local manager and payload release control, tasked by the UAV manager on Impavida over MQTT at 5.8 GHz	Custom	In-house
Open source software	ArduPilot, Ultralytics, OpenCV	ArduCopter 4.7, pymavlink, Ultralytics YOLO, OpenCV, MQTT and Protocol Buffers, Mission Planner	Open source	0
Other: embedded	Seeed, Waveshare	XIAO ESP32-C6 (6) with antennas, for communication routing; RP2040-Zero (5) for the kill units; Raspberry Pi 5 supplies (2); inline fuses	Purchased	97.11
Total				9,605.45

APPENDIX C: TASK SEQUENCES

Full sequences per task and tier, as designed and, where a simulator or trial exists, as exercised.

1) Task 1: Safe Passage

Core tier. Impavida identifies and reports the markers, circles the entry buoy clockwise, transits with flashing RED to starboard and GREEN to port while avoiding unlit obstacles, and circles the exit buoy counterclockwise [1]. The perception and guidance chain is:

1. The vision computer detects a buoy and classifies the beacon color and flash state over successive frames against the one-second flash cycle.
2. Range to the detection is estimated from stereo depth. Combined with own pose at the frame timestamp, this yields the buoy coordinate in the course frame. A distant buoy that yields a bearing but no range is held as a ray track and becomes a search lead, not a navigation target.
3. Coordinate and color are written to the map layer, where they are fused into landmark tracks.
4. Once sufficient classified tracks exist, the path planner produces a reference path. Because the exit point is not known in advance, the path is generated gate by gate, with two, one or zero gates of look-ahead and with approach heading corrected by gate color.
5. The controller computes cross-track error against the nearest point on the reference path and follows it with line-of-sight guidance.

When the plan runs empty, the search state machine runs leads, outward spiral and frontier in that order, each phase ending the moment a usable waypoint appears.

LiDAR is deliberately excluded from the Task 1 perception chain. A buoy at gate-detection range returns only a handful of points, and on a hull moving in waves the LiDAR must run in its fast scan mode, after which motion filtering leaves so few true returns on the buoy that they cannot be separated from noise. Stereo vision, which classifies on color and shape rather than on point count, is the right sensor for small floating targets.

Advanced tier. The safe route is indicated by beacons visible only from above, so the USV cannot resolve the gates by itself. Buzzlight overflies the field, identifies the gate beacons, and reports the route to the Global Mission Supervisor on Impavida, which enters the gate positions in the map layer and forwards the route to RoboCommand through the OCS. Gate detection runs on Buzzlight's own computer, so only gate coordinates cross the air link, never imagery, and a dropped link delays the route rather than losing it.

Disruptive tier. The safe route may change during transit, and the change is visible only from above. Buzzlight remains on station and pushes map updates in real time. This tier is attempted within the UAV's battery endurance. Whether the endurance covers the run depends on course duration, which in turn depends on the path planner producing the route continuously, without search pauses, so the Disruptive attempt is tied to the performance of the gate-by-gate planner rather than to a fixed go or no-go.

2) Task 2: Infrastructure Survey and Repair

Core tier. Impavida opens the task. Its depth camera recognizes the active entry buoy (GREEN indicator), establishes its coordinates, and transmits them to Riptide over the acoustic modem. Riptide acknowledges receipt and begins its transit immediately. The acknowledgement is what tells the Global Mission Supervisor that the vehicle is under way.

Riptide transits on the surface to retain GNSS, using a receiver mounted in an acrylic enclosure on top of the watertight tube. Two to three meters short of the buoy it submerges. GNSS and the RF kill-

switch link are lost at this point, so the descent is a controlled vertical submerge on the heave thrusters, holding attitude, rather than a pitched dive, which would induce forward motion and cross-flow drag.

Submerged, the four-element hydrophone array gives bearing to the acoustic pinger suspended below the active buoy, and the vehicle homes on it. The pinger marks the buoy, not the pipeline, so from the buoy the vehicle scans a full turn until at least three ring lights form a cluster in the camera image.

The mission controller is a finite state machine of seventeen states (Figure 5). After the vertical submerge it checks height above the pipeline, aligns the camera's center ray with the light cluster, and follows the pipeline in short forward bursts with yaw servoing on the cluster, recording each ring as intact (GREEN) or damaged (RED) and logging the pinger channels as it goes. Ranging uses the projected area of the ring in the image as a distance proxy, with frame skipping. When the last light leaves the image it turns through 180 deg, returns to the RED ring, shifts laterally to bring the pipe axis horizontal in the image, centers the ring, advances until the ring fills a set fraction of the frame, holds the magnetic probe against it until the camera registers the change to GREEN, decodes the flash pattern, and surfaces on upward velocity. In the mission simulator, which supplies ring colors as metadata, this sequence completed 100 of 100 runs across three scene types and twenty initial headings. The report order of the segments comes from the pinger: when a ring sits inside a fixed image window at the right apparent size, the pinger intensity at that moment is recorded, and since the single pinger hangs at the start of the pipeline, intensity falls along it and the sequence orders the rings. Once the nearest ring is identified RED the remainder report GREEN without separate classification. Ordering by vehicle position is kept as the fallback, because shallow-water multipath can make intensity non-monotonic over a few meters. Throughout the underwater stage a heartbeat and task-status message goes to Impavida over the acoustic link. The pipeline status report, ordered from the segment nearest the active buoy, is sent on completion, and surfacing to report is permitted [1]. The first Core objective is the survey and the ordered report. The repair earns additional credit and is the prerequisite for the higher-tier delivery sequence [1].

Three constraints shaped this sequence. The acoustic transducer is mounted aft, so Riptide presents its stern to Impavida to hold the link. The probe arm must land the magnet without the vehicle striking the structure and without the arm obscuring the useful part of the camera field of view. The final approach is the leg that needs controller gain and actuator limit tuning in the water, and it is scheduled first when water time is available.

Advanced tier. After the repair the ring flashes one color indicating the delivery circle [1]. Riptide decodes the pattern, transmits the resource request to Buzzlight through the supervisor on Impavida, and reports task completion to RoboCommand. Buzzlight, already airborne in autonomous mode with a tin pre-loaded, acknowledges the re-tasking, transits to the delivery platform and releases the tin over the indicated circle [1]. The Disruptive extension, which requires autonomous retrieval of a specific tin from a platform, is not targeted.

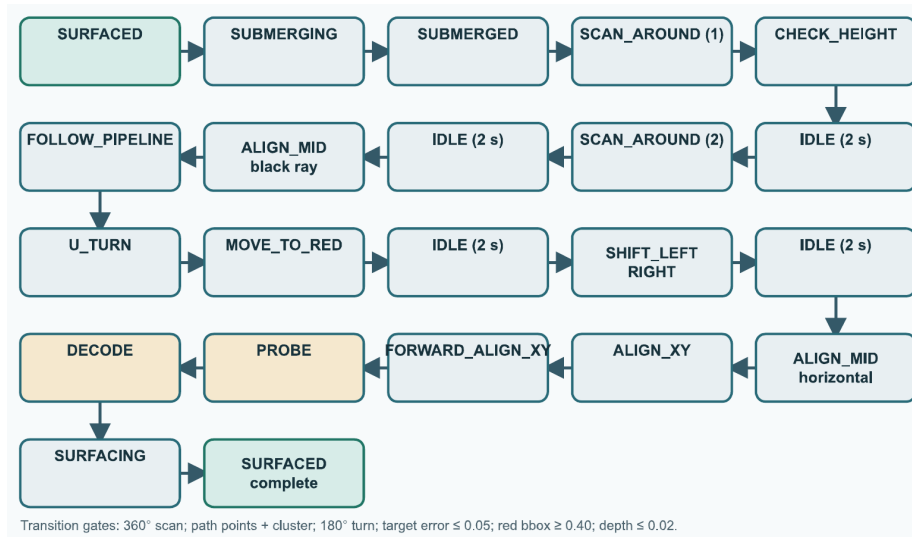


Figure 5. Riptide mission state machine, SURFACED through PROBE and DECODE to SURFACED complete. The cross-vehicle message exchange is described in the text.

3) Task 3: Coordinated Logistics

Core tier. Impavida locates the docking bay marked by a GREEN indicator, docks fully, reports to RoboCommand, locates the activated RED window, sprays water until the light turns GREEN, detects the five-second completion indication, and reports completion [1]. The sequence is:

1. Transit to the task starting position.
2. If no structure is recognized, rotate in place. This is the default search behavior.
3. Establish a line parallel to the docking front with the Task 1 method applied to the dock: the close-range camera ranges each of the three bays, the three positions go on the map, a line is fitted through them and offset at the required standoff to produce the sliding path (Figure 6). The same detector, map and planner serve both tasks. A LiDAR plane-extraction method for the same step was developed against the in-house point-cloud simulator and is held in reserve.
4. Sway along the offset line using the lateral thrusters under CUSTOM_AUTO, recording the indicator color and coordinate of each bay. All three bays are confirmed rather than stopping at the first apparent match. The side thrusters were installed specifically for this maneuver.
5. Return to the GREEN bay by remembered heading, compass or GNSS, re-confirm it by vision, and engage a slow controlled approach with thrust cut at a critical standoff distance.
6. Report the docking. The RED window light activates. The same close-range camera localizes the window and passes the target to the water-cannon manager, which aims the cannon and sprays until the camera registers the change to GREEN, then reports completion. The cannon is a 12 V R385 diaphragm pump feeding a 3D-printed PETG nozzle with a 4 mm exit, carried on a servo-driven PETG mount (Figure 7), with the pump and servos powered from a 6S pack through a buck converter and switched by the cannon's own Raspberry Pi.

The task state machine includes a wrong-bay recovery branch that undocks the vehicle and returns it to the sweep.

Advanced tier. After the window turns GREEN the light flashes one color for 60 s [1]. Impavida decodes the pattern, relays the resource request to Buzzlight and to RoboCommand, and Buzzlight

delivers its pre-loaded tin to the indicated circle. Impavida need not remain docked during the delivery [1]. As in Task 2, the Disruptive extension is not targeted.

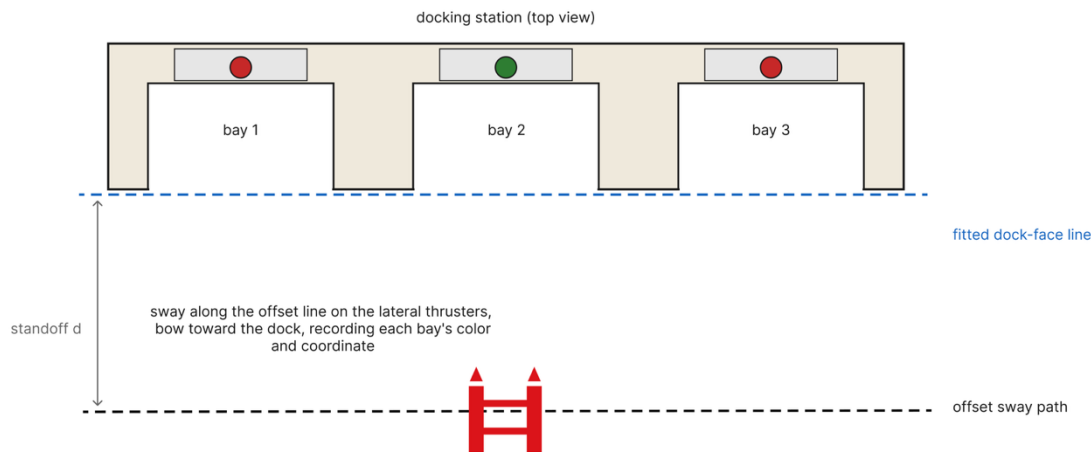


Figure 6. Docking search: fitted dock-face line, offset sway path, bay confirmation, return and slow approach.

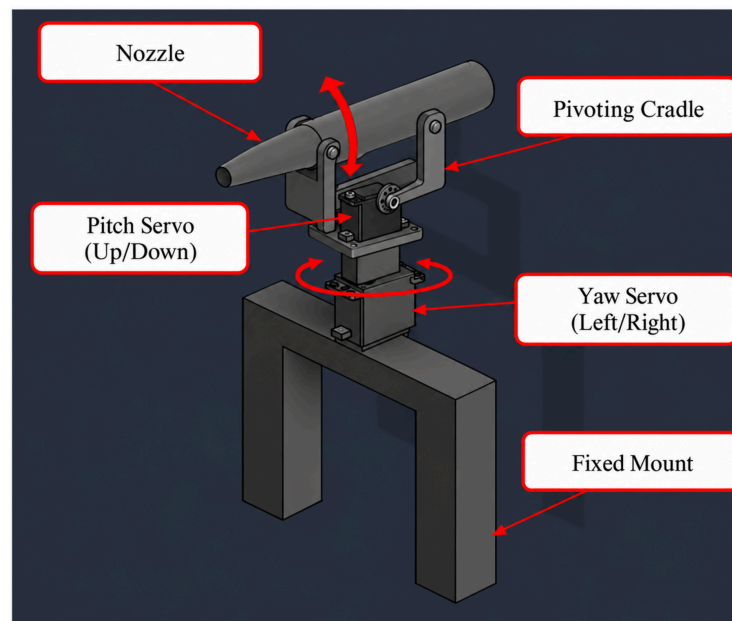


Figure 7. Water cannon assembly.

4) Task 4: Dynamic Incident Response

Core tier. RoboCommand transmits a request for assistance at a GPS position for a specified domain. The tasked vehicle acknowledges, breaks off its current task, transits to the position, loiters, reports readiness, and resumes after RoboCommand clearance [1]. The mission executive treats an incident as an overlay on the running task rather than as a replacement for it: the current task state is preserved, the incident sequence runs to completion, and the task resumes from where it was interrupted. The local manager gives the assistance request priority over the running task, and the request propagates from the

OCS through the supervisor to every module of the control architecture so that planning, guidance and reporting act on the same tasking state.

Advanced tier. RoboCommand transmits a static keep-out area. Vehicles in the affected domain acknowledge and avoid it until an all-clear is received. The area enters the map layer as an exclusion zone, a planner input rather than a special case in the controller.

Disruptive tier. RoboCommand reports a moving surface object with position, heading, speed and type. Vehicles keep at least 10 m clear based on current and projected travel [1]. This requires re-planning against a time-varying exclusion zone, which the same path planner handles as a moving obstacle with the obstacle-aware fallback engaged. All three tiers of Task 4 are targeted, because each is an extension of the same executive and planner rather than a new subsystem.

APPENDIX D: PERCEPTION DETAIL

Training, validation, depth sampling, the detection-to-map transformation and the LiDAR pipeline held in reserve.

Interface. The vision computer returns the coordinate and color of detected objects. It never returns imagery. Own pose is pushed to the vision board over the vehicle Ethernet and used to stamp each processed frame, and the return may carry a frame index rather than a timestamp, so that repeated observations of the same object can be checked for positional drift. This keeps inter-board bandwidth low, prevents perception latency from propagating into the control loop, and makes the map layer independent of the perception implementation.

Two stereo cameras. Impavida carries two ZED stereo cameras [27] with different baselines and focal lengths, and the task-specific recognition module selects between them by mission state. The ZED 2i with 4 mm lenses is the far-range unit for Task 1 buoy detection. The ZED Mini is the close-range unit for Task 3, used for the docking line estimate and for aiming the water cannon, where a wide field of view and near-field depth accuracy matter more than reach.

The far-range choice was fixed by simulation rather than by datasheet reading. In a sweep of 1800 course layouts across the rules' range of turn angle, gate spacing and gate width, the 4 mm unit (72 deg field of view, about 25 m of usable depth range, 40 m bearing-only) handled every turn to 16 m gate spacing and most to 24 m, losing only at 30 m, while the original 2.1 mm unit (110 deg, 12 m, 20 m) was spacing-limited beyond 16 m. In a second sweep of 540 runs over depth range, bearing-only band and field of view, nothing worked below 20 m of depth range on the hardest course at any field of view, and moving from 12 m to 25 m of depth range bought more than any tuning did anywhere. The operational limit of Task 1 is a camera limit, not a tuning limit, which is what justified a second camera rather than a longer parameter search.

Camera and LiDAR. The two sensor types were assigned to distinct problems rather than fused.

TABLE D1
SENSOR ASSIGNMENT ON IMPAVIDA

	Camera (ZED 2i 4 mm, ZED Mini)	LiDAR (Livox Mid-360)
Provides	Rich features, color, high-confidence classification, stereo range	Geometry as a point cloud, no color
Weakness	Cannot reliably extract a plane from a partially visible structure	Small floating targets return too few points at range under platform motion
Assigned to	Object classification, color and flash-state discrimination, pipeline tracing, dock-line estimate, final approach, cannon aiming	Dock-line extraction, developed in simulation and held in reserve

The surface detector is a team-trained Ultralytics YOLO network [4] on the perception computer: a YOLO11n fine-tune on traffic cones for the development and land tests, and a YOLO26-L on the course targets listed below for the competition. The cone model was fine-tuned at 640 px on 399 training frames from indoor and pool video, with cone cutouts composited onto pool, indoor and noise backgrounds, red and green non-cone distractors and negative-only images included so that the network cannot rely on color alone. On 14 held-out frames from familiar scenes it scored 98.9 % precision, 100 % recall and 99.5 % mAP at 0.5 IoU. An earlier indoor-only pilot scored 49.1 % mAP on a transfer test with green cones the weakest, which is what drove the scene-diverse compositing.

Range is taken from the stereo depth inside each detection rather than from the whole box: the depth code samples the central 30 to 70 % of the box width and 35 to 80 % of its height, keeps only pixels that pass a class-specific HSV mask and have finite positive depth, requires at least 12 valid points and a 25 % valid fraction, rejects outliers by median absolute deviation, and reports the median position of what

remains. Against tape-measured cones over 11 readings from 5 to 35 m, camera range agreed with true distance to a mean absolute deviation of 0.41 m, with readings above 30 m stable in value but flickering in detection. That boundary is the 25 to 30 m usable depth range the simulator sweeps assumed. The result carries one caveat until the over-water sweep runs: stereo matching depends on scene texture, and water is low-texture, specular and moving, unlike the hard standing the cones stood on. Landmark tracks on the map keep a decaying weighted average of the last ten detections and expire after 2 s unseen.

From detection to map coordinate. Each accepted detection is converted in four steps. Rectified stereo depth is

$$Z = f_x B / d, \quad (1)$$

with f_x the focal length in pixels, B the baseline and d the disparity, so a fixed disparity error δd gives $\delta Z \approx Z^2 \delta d / (f_x B)$. That quadratic growth is the $c r^2$ term in the simulator's range-noise law, and a longer focal length is the only lever on it, which is the reason behind the 4 mm lens. The retained points inside the sampling region give the median camera-frame position $(\hat{X}, \hat{Y}, \hat{Z})$ and the range

$$r = \sqrt{\hat{X}^2 + \hat{Y}^2 + \hat{Z}^2}. \quad (2)$$

With camera mounting yaw α and forward and right offsets t_F, t_R , the vehicle-frame forward and right coordinates are

$$F = t_F + \hat{Z} \cos \alpha - \hat{X} \sin \alpha, \quad R = t_R + \hat{Z} \sin \alpha + \hat{X} \cos \alpha, \quad (3)$$

and with vehicle position (N_v, E_v) and yaw ψ clockwise from North at the frame time, the map coordinate is

$$N_c = N_v + F \cos \psi - R \sin \psi, \quad E_c = E_v + F \sin \psi + R \cos \psi. \quad (4)$$

An observation is accepted when the detector confidence is at least 0.40, the valid-depth fraction q is at least 0.25, the range lies within the configured cap (20 m in the 22 September test) and the pose used is no older than 0.5 s. It is associated with the nearest track of the same class within 3.0 m, tracks unseen for 2 s are dropped, and the displayed track position is the decaying average over the last ten detections,

$$\tilde{N} = \sum_a w_a N_a / \sum_a w_a, \quad w_a = 0.92^a, \quad (5)$$

with a the sample age in detections. Roll and pitch are not compensated in (3) and (4), and supplying (N_v, E_v, ψ) at the frame time rather than at processing time is the observer's pose-history function. The displayed spread of a track is dispersion about its own estimate, not error against a surveyed position. Two cones that are separated in the image resolve as two landmarks. When one cone almost fully occludes another the two can merge into one, which is carried as an operating constraint rather than a defect. Beacon flash state is resolved by observing each detection over successive frames against the one-second flash cycle defined in the rules, with at least eight frames spanning one full cycle before a classification is committed. A polarizing filter is fitted to the far-range camera to suppress surface glare.

LiDAR, in reserve. A LiDAR detection pipeline exists and runs end to end on live Mid-360 data: RANSAC plane segmentation removes the dominant flat surface, which on the water is the surface itself, Euclidean clustering isolates the objects above it, and a persistent tracker assigns stable identities with exponential smoothing on position and size to suppress the jitter of the sensor's non-repetitive scan pattern. It was built twice, in Python with Open3D for tuning and in C++ with PCL as a ROS 2 Humble package, unit-tested against synthetic point clouds before each live session, and validated indoors on the Jetson with RViz2. Because the sensor returns range and intensity only, it cannot read the colored indicators the course tasks turn on, and its shape-only classification was not validated on course targets before the deadline, which is why it stays in reserve.

Underwater. Riptide carries a Blue Robotics low-light USB camera on a servo mount and four Aquarian Audio S1n hydrophones with PA4 preamplifiers into a Measurement Computing USB-1608FS-Plus acquisition board sampling at 100 kS/s. The array gives bearing to the pinger from the

squared-range differences across its four elements. A compact YOLO11n detector finds the ring lights. It was trained on the UW-Light-Detection Benchmark [5], [6], 2,200 underwater images of light sources split 1,540, 330 and 330, as a single light-spot class for 100 epochs at 640 px (Figure 8). On the 330 held-out validation images it reaches 99.1 % precision, 98.9 % recall, 99.4 % mAP at 0.5 IoU and 82.4 % mAP at 0.5 to 0.95, from a 5.5 MB checkpoint, and runs at 13 ms per frame on a laptop GPU with no loss of accuracy in half precision. The deployment target is TensorRT half precision on the Orin NX, for which the published YOLO11n reference is under 5 ms per frame [4]. INT8 is not used, because the published INT8 accuracy on that board collapses without dataset-specific calibration. Ring color is classified by an HSV mask inside each detected box, and ranging uses the projected blob area of the ring as a distance proxy, with frame skipping.

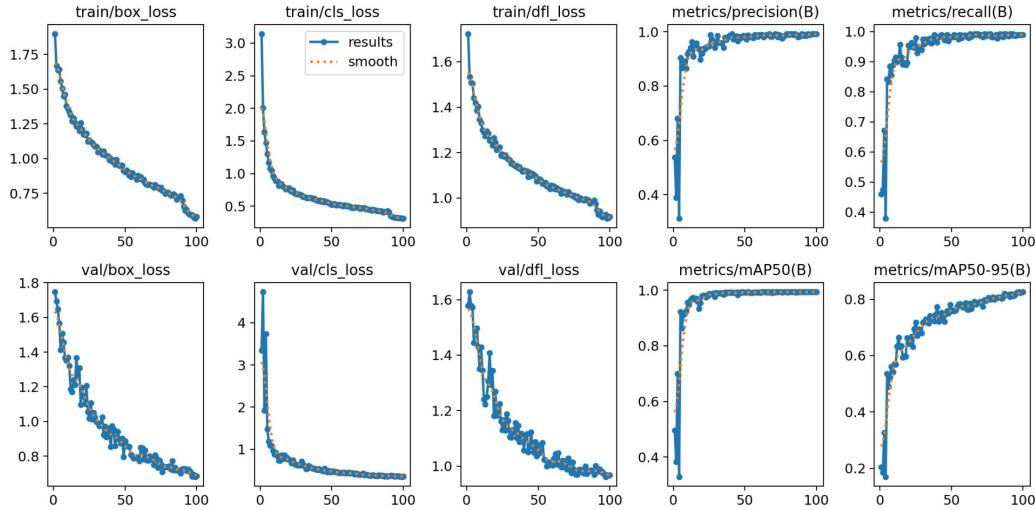


Figure 8. YOLO11n training curves.

Air. Buzzlight carries a Blue Robotics low-light USB camera, the same unit as on Riptide, and runs all of its own perception on its Jetson Orin NX: a YOLO11-L detector with color detection for the overhead beacons, gates, landing pad and delivery circle, with only coordinates sent to the boat. Camera-assisted return to land against a printed target is the backup to GNSS return-to-launch under drift.

Training targets. Buoys (Task 1), pipeline beacon clusters (Task 2), docking structures (Task 3), the 2 m by 2 m landing platform and the delivery platform (Tasks 2 and 3, UAV).

Software: ZED SDK for camera handling [27], OpenCV for image processing and object recognition [28].

APPENDIX E: SIMULATION STUDIES

The map layer and planner detail, and the simulators and studies.

Map layer. The map layer is a distinct module addressable by both the controller and the supervisor. It holds the absolute map with detected objects anchored to it as landmark tracks and generates the reference path. Tracks carry a 2 by 2 position covariance, are associated by a Mahalanobis gate, and are fused by inverse variance with process noise so that the estimate follows the EKF's own drift. Color is classified over successive frames by the unlit fraction of the observations. Only classified tracks with position uncertainty below 1.5 m are planned on. Bearing-only detections are held as ray tracks that never update a position until a depth detection lands on their bearing. Everything still unclassified is a lead for the search. The distinction between path and trajectory is deliberate: the map layer produces a geometric path with no velocity or timing component, and timing is the controller's concern. The map is deterministic in the first implementation. Exclusion zones from Task 4 enter it as constraints.

Gate structure. Gates are red-green pairs with an intrinsic crossing direction, green to port and red to starboard. Items are chained greedily from the boat with penalties for items behind the vehicle and for backward crossings, with 3 m of hysteresis so the first item does not flip. Each item receives pre, cross and post waypoints at 3 m along its direction so that the crossing is perpendicular. The boat may commit to a crossing only from inside an approach corridor, and once committed it follows the gate axis by cross-track LOS. The exit is appended to the plan only once no unresolved leads remain in the search radius.

Path planner selection. USV path planning has a large literature and few field-validated planners [8]. The team benchmarked six candidates against the same pose, target and known obstacle map, each planning only to the currently available objective because the exit is unknown: Dubins paths [9] with rotate-in-place as the baseline, A* [10], Hybrid A* [11], D* Lite [12], RRT* [13], and a visibility graph with A* [14]. Thirty course seeds per condition were run with exact geometry, so the planner was isolated from perception.

Collision-free path rate. Median solve time in the full-stress microbenchmark, relative values on the same machine: Dubins 0.04 ms, Hybrid A* 0.18 ms, A* 5.2 ms, D* Lite 5.6 ms, RRT* 23.9 ms, visibility with A* 35.6 ms. Hybrid A* paths were about 57 % distance-efficient in the sharp-turn test against roughly 99 % for the grid and graph planners.

The decision that follows is the one in, item 4. Dubins with rotate-in-place remains the normal planner because it is deterministic, two orders of magnitude cheaper, and exploits a differential-thrust hull's ability to yaw on the spot. An obstacle-aware planner, A* or D* Lite on the deterministic map, is engaged only when the direct route fails the clearance check. Gate-transition runout scales with the required heading change, speed is reduced near difficult transitions, and for severe turns the vehicle clears the gate, yaws on the spot and proceeds rather than following a wide loop. The path is regenerated as new tracks enter the map.

Riptide's pipeline following runs in the image plane rather than on a map (Figure 9). The centers of the detected ring lights are summarized by their mean and principal axis. A ray from the image center to the mean sets the yaw target, a perpendicular through the cluster center sets the lateral follow direction, and the point where that perpendicular meets the image edge is the follow target. Alignment closes normalized image errors with proportional gains and a small tolerance, and lateral shifts are commanded from the sign of the arrow rather than from any world-frame estimate of which side of the pipe the vehicle is on. This keeps the underwater loop free of the USV's map and estimator, which the acoustic link could not feed at a useful rate.

Search when the next gate is not observable. A stationary 360 deg scan cannot solve a range problem: after the scan the boat is exactly as far from the gate as before. The search therefore investigates bearing-only leads first, then runs a heading-aligned outward spiral that starts at the boat's position along its current heading and expands until a usable target appears, then falls back to frontier exploration. In

the kinematic simulator, investigating leads before scanning cut search time to about a third on narrow-field-of-view U-turn courses, and over 15 seeds at Hard difficulty stationary scanning completed 66.7 % of runs against 86.7 % for the spiral.

Kinematic simulator. The Task 1 chain was developed in a purpose-built top-down simulator that models the problem at the detection level rather than the physics level. The camera reports each buoy in view with a stereo range whose noise grows as $a + b r + c r^2$, with $c = 0.0015 \text{ m/m}^2$ from the ZED datasheet accuracy bound [27], a bearing with 0.25 deg noise, and color and miss probabilities that rise toward the edge of range. Beyond the usable depth range the detector still reports bearing and color without range. Lights flash at the rules' one-second cycle with random phase. The boat's own position is an EKF-style solution with Gauss-Markov drift (0.5 m, 60 s in position and 1.5 deg, 30 s in heading), and everything the boat computes lives in that drifting frame while only scoring uses truth. Courses are generated from a seed with controllable turn angle, gate spacing and gate width, so a run is repeatable and a sweep is possible. The hull is a kinematic unicycle with direct velocity command: hull hydrodynamics are not modeled in this module, so there is no inertia, added mass, damping, sideslip, or wave, wind and current forcing. The module tests how noisy detections become a route, not how the hull follows one. Hull dynamics enter at the Gazebo stage and in the water. The perception model also omits false positives, occlusion, detector latency and roll and pitch coupling, so the module's success rates are upper bounds.

TABLE E1
SIMULATION STUDIES AND FINDINGS

Study	Runs	Finding
Nine-knob tuning sweep (Latin hypercube, 3 course scenarios by 4 seeds)	780	The gate-approach corridor half-width is the decisive parameter: success collapses below 0.4 m under 0.5 m of EKF drift and plateaus from 0.8 m. Best configuration 75 % strict and 100 % task success at 75 s P90 mission time, against 58 % and 83 % for the defaults. Look-ahead, sweep angle and frontier weights are flat within their ranges
Course-layout sweep (turn by spacing by width, two cameras, three tunings)	1800	Operational limits are camera limits, not tuning limits. Gate width from 1.8 to 3.2 m is flat for both cameras. Tuning moves the edges by one grid step at most
Camera-range sweep (depth range by bearing-only band by field of view, three courses)	540	Nothing works below 20 m of depth range on the hardest course at any field of view. Depth range dominates field of view
Planner benchmark, six planners (student extension)	30 seeds by 3 conditions	Every planner reaches 100 % in open geometry; obstacle-aware planners hold 97 to 99 % under sharp turns and obstacles where Dubins alone falls to 73 % and 44 %
Search strategies, three variants (student extension)	15 seeds by 2 difficulties	Spiral search completes 86.7 % of Hard runs against 66.7 % for stationary scanning

Four seeds per cell put the rates at about plus or minus 15 %. The ordering of the findings is robust to that. The exact numbers are not, and none of them is a prediction for the water. Three design changes came out of these runs and are recorded in.

Riptide mission simulator. The underwater mission was developed in a second purpose-built simulator. A Python mission loop at 10 Hz reads vehicle state, camera detections, four hydrophone channels and path geometry over a local HTTP interface, runs the state machine and the yaw, forward, lateral and vertical controllers, and posts vehicle commands back. Hydrophone ranges are sampled at four element positions, bow and stern 0.36 m apart and port and starboard 0.24 m apart, so the bearing law runs on the same geometry as the physical array. The simulator supplies ring colors as metadata, so the detector is not in this loop, and the hull is kinematic. A 197 s recorded demonstration covers every reachable state (Figure 10), and a 100-run stress record on 18 September completed 100 of 100 missions

across default, easy and randomized scenes with twenty initial headings, at a mean completion time of 87 s. This supports the sequence logic and nothing about the physical vehicle.

Gazebo. In parallel with hardware preparation and vehicle assembly, the team ran simulation in Gazebo Harmonic [25], the maritime simulation lineage of the RobotX virtual competition [26], with the ArduPilot software-in-the-loop autopilot and the flight controller in the loop. The model currently contains Impavida and Buzzlight. It was used in August to validate the line-of-sight controller before the first autonomous water trial, and it is used with the flight controller in the loop to configure and check ArduPilot parameters against the vehicle model before they are loaded on the boat. The next use is separate mission stages run end to end. The in-house LiDAR simulator allowed the dock surface-extraction algorithm to be developed against noisy synthetic point clouds before the sensor was on the vehicle.

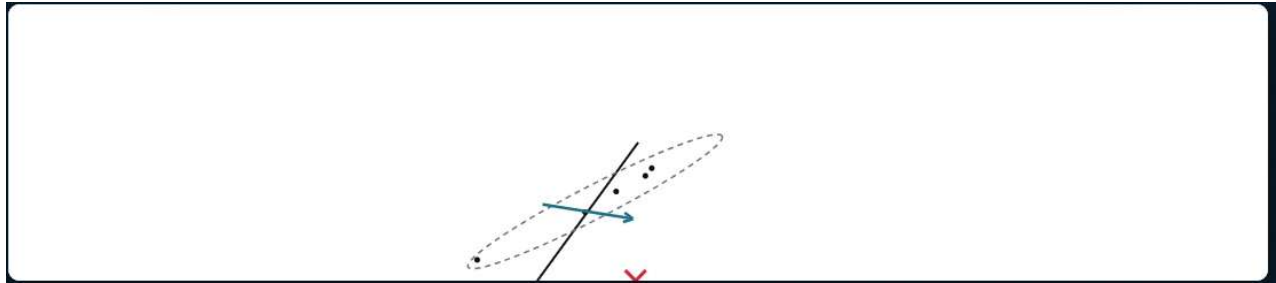


Figure 9. Image-plane path generation view.

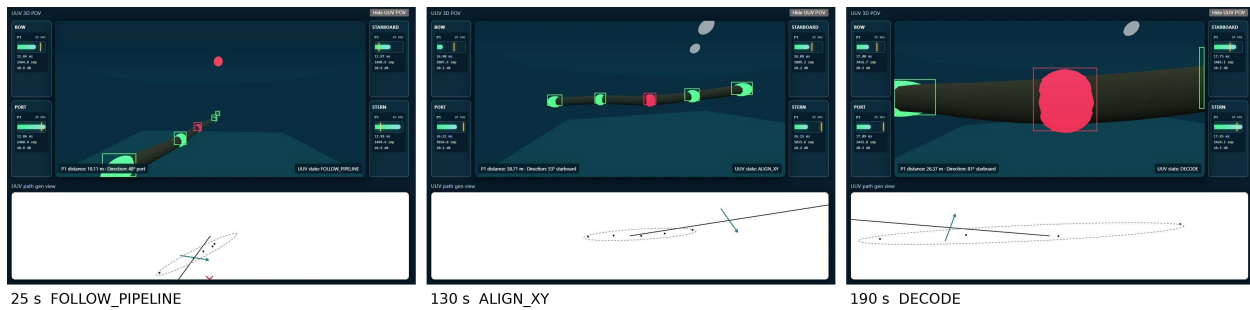


Figure 10. Riptide mission simulator frames at FOLLOW_PIPELINE, ALIGN_XY and DECODE.

APPENDIX F: COMMUNICATIONS DETAIL

Link table, conventions and the acoustic link model.

Topology. The team network has a ground segment and a vehicle segment (Figure 11). On the ground, a router is the hub. Attached to it by Ethernet are the OCS, one ground station per vehicle, and a 5 GHz radio feeding a 21 dBi, 60 deg sector antenna. Every vehicle has up to four radio paths of different kinds: a primary IP link at 5.8 GHz, a secondary MAVLink telemetry link at 915 MHz to the vehicle's own ground station, a safety link with an RP2040 controller at each end, and a manual RC link. Impavida holds the primary IP link to the ground and a second 5.8 GHz radio that serves as the IP link to Buzzlight, so the UAV's mission traffic goes through the boat and its own path to the ground is its telemetry link. Riptide reaches the boat by the acoustic modem when submerged and by a 915 MHz radio when surfaced.

TABLE F1
RADIO AND ACOUSTIC PATHS

Path	Radio	Band	Ends
Primary IP, surface	UISP airMAX Bullet with 6 dBi omni; ground: Rocket AC Lite with 21 dBi sector	5.8 GHz	Impavida to ground router
Primary IP, air	UISP airMAX Bullet with 6 dBi omni; boat: Rocket AC Lite with 13 dBi omni	5.8 GHz	Buzzlight to Impavida
Secondary telemetry, each vehicle	Holybro SiK Telemetry V3 pair	915 MHz	Vehicle to its ground station
Safety, Impavida and Riptide	Holybro SiK Telemetry V3 pair with RP2040 operator and vehicle units	915 MHz	Kill operator to vehicle
Manual RC, Impavida	Radiomaster GX12 to Holybro Zephyr ELRS Gemini X, CRSF; BlueBoat BaseStation with Xbox controller as the second manual path	2.4 GHz and 868 MHz; 2.4 GHz Wi-Fi	Operator to flight controller
Manual RC, Buzzlight	FrSky X20S to FrSky TD10, SBUS	2.4 GHz and 868 MHz	Operator to flight controller
Submerged	Water Linked M16 acoustic modem	58.5 kHz	Impavida to Riptide
Ground LAN	UISP router	Ethernet	OCS, three ground stations, sector radio

Small XIAO ESP32-C6 modules handle communication routing between the vehicle links and the onboard network. Every radio is declared on the RF emitter template with its configured channel set [1]. The OCS detects loss of a vehicle link by heartbeat timeout, and the kill link acts independently of the mission network.

Message layer. MQTT [19] is the inter-vehicle transport, with Protocol Buffers payloads [20]. MQTT also carries requests between the control and compute layers within a vehicle. Message schemas are defined independently of the transport, so the wire format remains reversible. ROS 2 was deferred as the inter-vehicle transport until hardware bring-up is complete. It is used only inside the LiDAR node set. The schemas follow the RoboCommand interface control document [21], with its four topic families (course, request, command, report) and the run-start sequence, so that the same message set serves internal coordination and judge-facing reporting [1].

Conventions. Every module and every message crossing a vehicle boundary follows one set of conventions, fixed before work was distributed and versioned from the first schema.

TABLE F2
CROSS-CUTTING CONVENTIONS

Item	Convention
Frames	Geodetic (WGS-84) across vehicle boundaries and to the OCS; local NED with an origin declared once per run for all navigation and control; body x forward, y starboard, z down; sensor frames as fixed offsets in one configuration file
Units	SI throughout; degrees and centidegrees only at the MAVLink boundary, converted once
Time	One UTC clock disciplined from GNSS; every message carries t_acquired (when the measurement happened) and t_published (when the module emitted it)
Identity	track_id (same physical object), detection_id (same message), seq (monotonic per stream)
Uncertainty	Covariance or a standard deviation with every pose, a confidence with every classification, and the error model with every range from blob area or signal intensity
Envelope	schema name and version, vehicle_id, seq, t_acquired, t_published, frame

Two of these were validated by the land test of 22 September: the missing acquisition stamp was the pose-interpolation error, and the declared origin is the startup check.

Acoustic link sizing. The submerged link was sized with the passive sonar equation [22] rather than by trusting the vendor's rated range. For the M16 at 58.5 kHz, transmission loss is $TL(R) = 20 \log_{10} R + \alpha R$, with α about 17.5 dB/km in seawater at 15 C and 35 ppt [23] and about 1 dB/km in fresh water. Relative to a 50 m reference the additional loss is +7.7 dB at 100 m and +15.4 dB at 200 m, which brackets the largest Impavida to Riptide separation in Task 2.

TABLE F3
ACOUSTIC TRANSMISSION LOSS RELATIVE TO A 50 M REFERENCE

Range (m)	Spreading, dB	Absorption, dB	Loss relative to 50 m, dB
50	34.0	0.9	0 (reference)
100	40.0	1.8	+7.7
150	43.5	2.6	+12.1
200	46.0	3.5	+15.4
300	49.5	5.3	+20.7

Fifteen decibels is small against the source level of a modem rated to several hundred meters, so range does not decide the link at course distances. What decides it is ambient noise at the venue, which at 60 kHz over the modem's receive band is on the order of 70 to 80 dB re 1 uPa for sea state 2 before harbor traffic [24], and shallow-water multipath, which no pool reproduces and which is also the one condition that can defeat the intensity-based segment ordering in Task 2. The design therefore keeps the acoustic exchange to one acknowledgement, presents the transducer toward the surface vehicle, and treats a venue noise measurement with the hydrophone array as the one input the model cannot supply. Modem parameters are those of the datasheet held by the team [29].

APPENDIX G: TRIALS RECORD

Bench, water and venue trial detail.

B. Bench tests

Each subsystem was bench tested first to verify the link and health of every component before integration: USV camera vision, LiDAR, the flight controller, and the communication network. The bench method is layered: one hardware or software layer is enabled at a time with an explicit pass or fail gate before the next is added. Lua probe scripts on the FCU and pymavlink STATUSTEXT listeners on the companion computer serve as diagnostic tools when the ground station software is not available. Pre-trial checks cover component security and both stops. For Riptide, buoyancy, trim, power isolation and controller feasibility are established on the bench before the vehicle enters the water.

C. Water trials

Water trials are organized per vehicle and per mission stage. Impavida completed flotation, manual-control and earlier autonomous navigation trials (Figure 12). The program for the remaining stages is ordered by the uncertainty each retires: the acoustic link with received level and packet loss versus range, the hydrophone array's bearing under rotation using a printed jig that holds the elements in as-installed geometry, camera field of view in air and then submerged, LED-ring mockups on the bottom for detection and ranging as a function of own depth, a closed-loop approach to a beacon on the Fathom ROV tether, Impavida's sway sweep past a mock dock face and slow approach with thrust cut at standoff, and a kill at depth to record Riptide's unaided ascent. Course element mockups follow the specifications published by RoboNation so that trials match competition geometry and colors: the docking box is built in PVC, the pipeline is taped pipe with a yellow base, spaced red bands and a green strip for the detector to train on, and underwater beacon mockups are LED rings on an independent battery, weighted for negative buoyancy, built to the enclosure geometry in the rules.

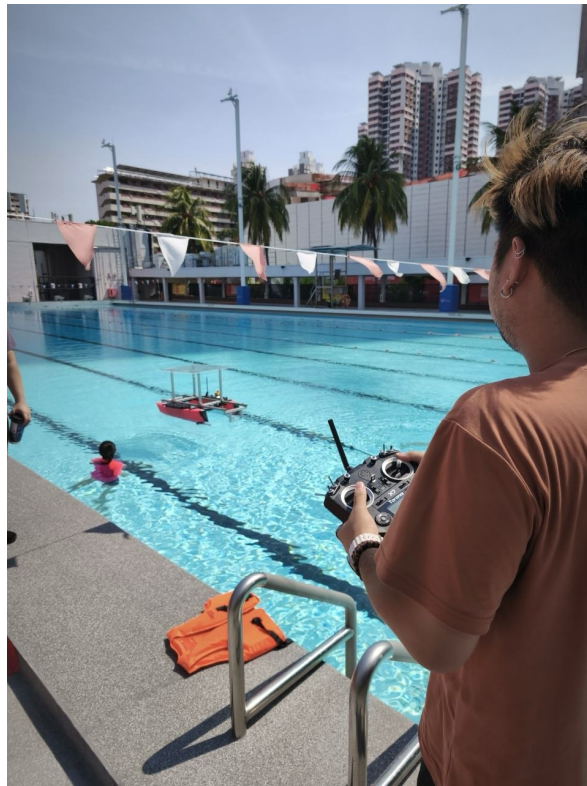


Figure 12. Impavida under manual control, 19 August.

Riptide's first water trials are tied to its Proof of Readiness, due 12 October, and are preceded by a GoPro visibility test at 10, 15 and 20 m in the trial water. Impavida's first pool session calibrates thrust straight and sideways at 10 to 40 % against GNSS, with the yaw moment of inertia taken from CAD with assigned masses before control tuning. For Buzzlight, loiter endurance with a tin loaded and drop accuracy over the target circle are ground tests that set the cradle geometry and release height.

D. Venue water

Local access to the competition venue is an advantage foreign teams do not have. Planned work in Marina Bay or Kallang water covers three things: a color wheel and LED mockups on a pole, filmed to calibrate color response under real light and water, a test of whether the pinger signal reflects usefully off the bottom, and an ambient noise recording with the hydrophone array, which is the one input the acoustic link model cannot supply. Open water at Coney Island is planned for GNSS accuracy and the effect of turbidity on underwater vision.

Eight findings from this period fed back into the design.

- In the August pool trials the companion-side controller, which had passed on the bench, did not hold its track in water, and the Proof of Readiness run used autopilot waypoint navigation instead. The suspected cause is a conflict between the heading setpoints commanded by the companion and the autopilot's own heading response in GUIDED mode. The corrective direction is to relax the heading constraint during line-of-sight transit and to stream position with velocity rather than bare position targets, and the hypothesis is tested in SITL before water: a straight path at 1, 2, 5 and 10 Hz with commanded against achieved velocity logged, the same with position-plus-velocity targets, then a look-ahead sweep at fixed rate recording cross-track error and the RMS of the thruster command derivative as a jerk proxy. The general lesson was that bench tests and kinematic simulation do not exercise the coupling between the companion's guidance and the autopilot's inner loop under real vehicle dynamics, and it is one reason the verification ladder treats water trials as a separate stage rather than a demonstration.

Bearing-only detections were first fused as points at an assumed range. Consecutive rays from a moving boat triangulated a confident ghost at the wrong range, which spawned a duplicate gate when real depth arrived. The fix, holding them as ray tracks that never update a position until a depth detection lands on their bearing, is now the design and was the only source of wrong-side passes in 780 runs.

- The first sighting of the solid-blue exit buoy ended runs with gates still unfound. The exit is now appended to the plan only once no unresolved leads remain in the search radius.
- The bare rulebook pairing window (6 to 10 ft) rejected real 1.8 m and 3.2 m gates under noise. A 0.5 m margin fixed it and the width sweep confirmed it.
- On 22 September the surface perception chain was run end to end on land with traffic cones standing in for buoys: the ZED 2i on the perception computer, the YOLO-class detector, and the observer taking attitude and position from the flight controller over MAVLink at 20 Hz, with each detection placed in the local North-East map through the yaw-rotated body-frame transformation. Over nine minutes the chain logged 4,504 accepted observations at about 5 Hz, at ranges from 1.6 to 20 m, and the saved map coordinates reproduced the transformation to numerical precision. The run exposed two things the design had anticipated and one it had not. Placing a detection with the yaw current at processing time, without interpolation to the frame time, is the error the observer's

pose-history buffer removes. A jump in the reported position at startup, before the origin settled, showed that the map must not accept detections until the origin is locked, which is now a startup check. And the navigation-quality fields were not in the log, so the run validated the chain and its timing, not the geographic accuracy of the map. The repeat on 23 and 24 September with the 4 mm unit logged the fix and satellite count (3D fix, 11 to 13 satellites), resolved two separated cones as two landmarks (Figure 13), mapped the occlusion boundary, which was the discriminating test agreed in supervisor review for the no-UAV Task 1 decision, and measured range against tape.



Figure 13. Two cones detected and mapped.

- The first cone model, trained on indoor frames only, transferred to the pool at 49.1 % mAP. The dataset was rebuilt with cutouts composited onto pool and indoor backgrounds, red and green distractors and negative frames, and the retrained model reached 99.5 % on held-out frames. The lesson, that the detector must not learn color alone, carries into the buoy model.
- A contrast-enhancement trial on the underwater detector went the wrong way. Applying CLAHE on the lightness channel to the same 330 validation images cost 1.4 points of recall and 8.3 points of mAP at 0.5 to 0.95 against raw frames, because the detector had been trained on the raw distribution. No contrast preprocessing is applied at inference, and any future enhancement is trained in rather than bolted on.
- An early reading of the rules placed the acoustic pinger at the pipeline start. On re-reading, the pinger is suspended below the entry buoy, which changed the Task 2 sequence and introduced the local search step from the buoy to the ring cluster.

APPENDIX H: VEHICLE DETAIL

Riptide power and data architectures, Buzzlight design iterations, and vehicle photographs.

Power (Figure 14). Two batteries feed two independent buses. Battery 1 feeds the propulsion bus through the kill-switch circuit and a SmartAP distribution board to the eight ESC and thruster channels, so a kill isolates propulsion and nothing else. Battery 2 feeds the electronics through a PM02D power module and a distribution board: the flight controller and its peripherals, the Ethernet switch, the Orin NX and the DVL on the 14.8 V rail, the four hydrophone preamplifiers and the subsea light on a regulated 9 V rail from a 12 A UBEC, and the mission computer with the modem and status indicator on a dedicated 5 V supply. Separating the buses keeps a propulsion kill from dropping the mission computer, so a killed vehicle keeps reporting through its ascent, and keeps thruster transients off the sensor rails, which matters for the hydrophone preamplifiers.

Data (Figure 15). Three compute elements, mirroring the USV partition:

TABLE H1
RIPTIDE COMPUTE PARTITION

Element	Assignment
Jetson Orin NX	All perception: computer vision, hydrophone array, DVL
Pixhawk 6X Pro	Low-level thruster and depth control through a Matek CAN-L4-PWM node, Bar30 pressure sensor, leak sensor, H-RTK ZED-F9P GNSS, subsea light and camera-mount servo outputs
Raspberry Pi 5 (16 GB)	Mission computer. The M16 acoustic modem terminates on its UART and the subsea LED indicator on a GPIO

The mission computer, flight controller, perception computer and A50 DVL share a GigaBlox Ethernet switch, and MAVLink between the mission computer and the flight controller runs over that switch rather than a serial line.

Underwater navigation is arranged around the fact that heading is the weak axis. The magnetometer sits inside a metal tube and does not read the earth's field, and the gyroscope drifts over slow turns, so computer vision is the relative reference submerged and pressure depth the absolute one, which is why the pipeline is followed in the image plane. The DVL gives velocity over ground in the body frame under bottom lock and can be fused with the IMU to bound position drift, but it does not fix heading, and the fusion stays an option rather than a plan until bottom lock at the working altitude has been checked in the water. The hydrophone array feeds the observer rather than the controller, since what it provides is a localization fix relative to the entry pinger.

Four further design features are on record. The GNSS receiver sits in an acrylic enclosure on top of the watertight tube, giving surfaced navigation without a separate mast. The acoustic modem is the primary link to Impavida, with a 915 MHz radio for the surface. The transducer is mounted aft. A magnetic probe is carried on an arm for the repair, and contact is confirmed by the camera when the ring changes from RED to GREEN.

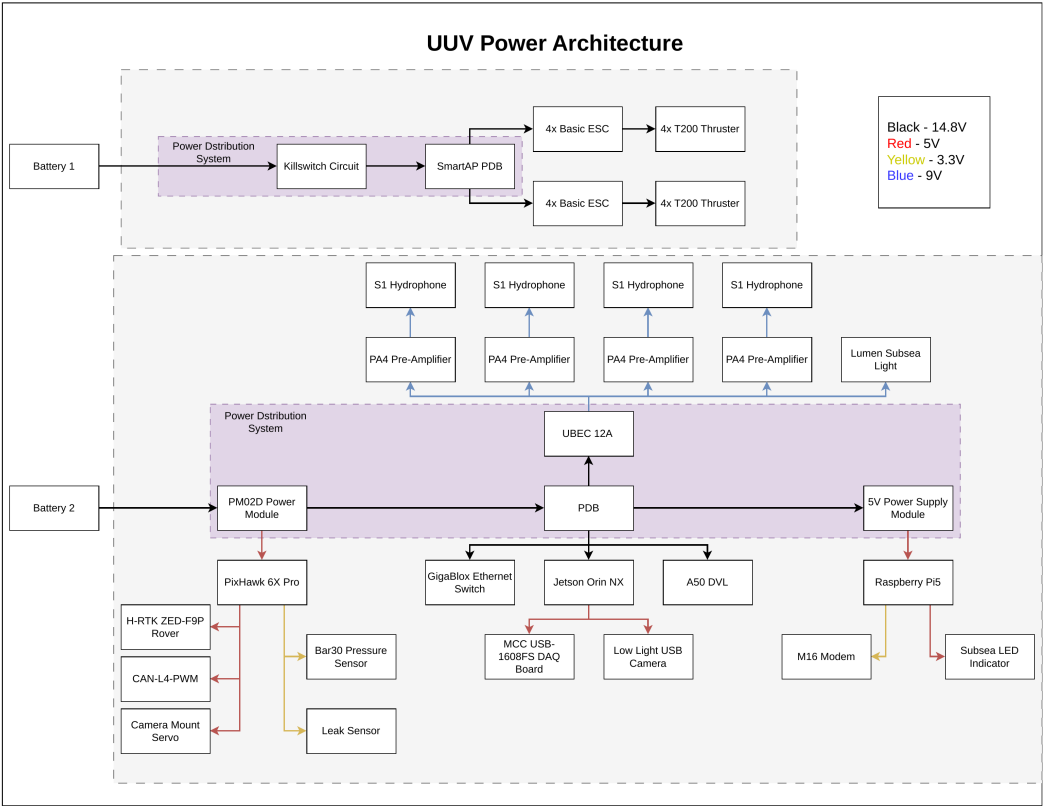


Figure 14. Riptide power architecture.

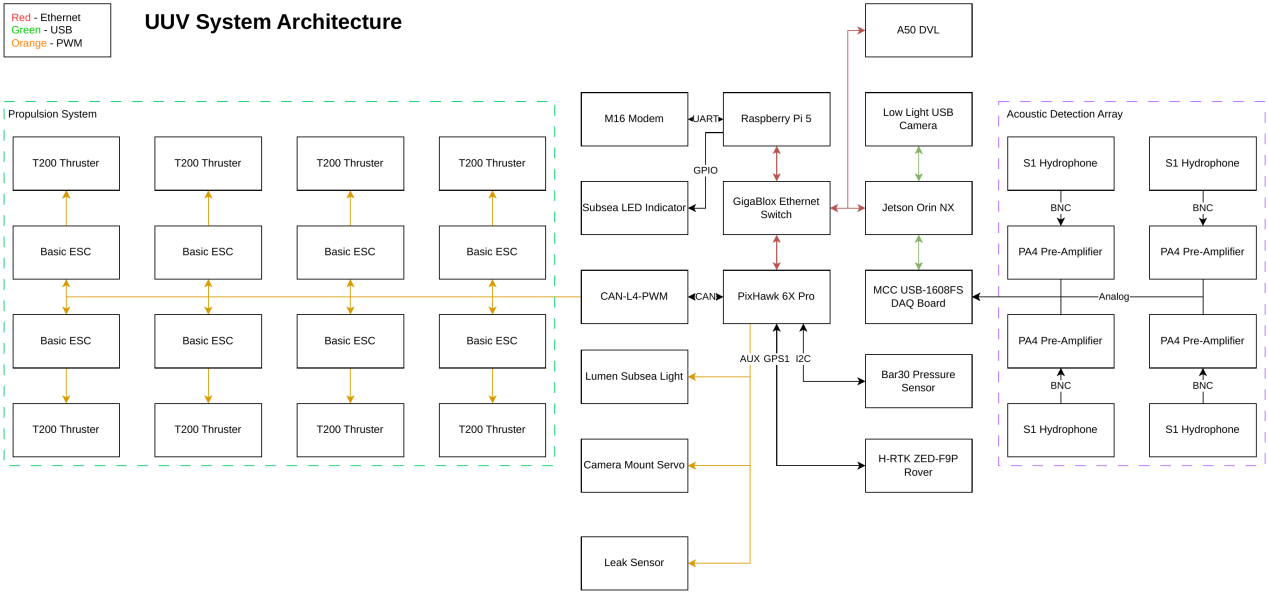


Figure 15. Riptide data architecture.

Two design iterations are worth recording (Figure 16). The battery was made modular so that packs can be swapped between the flight and delivery configurations. Following a center-of-gravity assessment, the enclosure was revised to remove the landing legs and integrate flotation for over-water operation. For the payload, a gripper and a passive-magnet interface were both considered. The magnet was ruled out by the aluminum tins, and the design converged on a mechanical release: the tin is pre-

loaded into a cradle and a small servo pushes it clear on command. The mechanism carries one moving part and no sensing beyond the servo command, which keeps its failure modes to release timing and drop accuracy over a 60 to 70 cm circle [1]. The cradle geometry and release height are set by drop tests on the ground.

Roles: overfly and gate reporting in Task 1 (Advanced and Disruptive), resource delivery in Tasks 2 and 3 (Advanced), and air-domain re-tasking in Task 4. Flight autonomy for delivery was assessed as straightforward (receive tasking, ascend, transit in guided mode, descend, release, return). The design risk is the communication path to the other vehicles, not the flight. Endurance is the limiting resource, and it sets the Task 1 Disruptive decision.



Figure 16. Buzzlight in floating enclosure.

Impavida CAD model is shown in Figure 17. Riptide CAD model is shown in Figure 18.



Figure 17. Impavida CAD render with the sensor frame fitted.

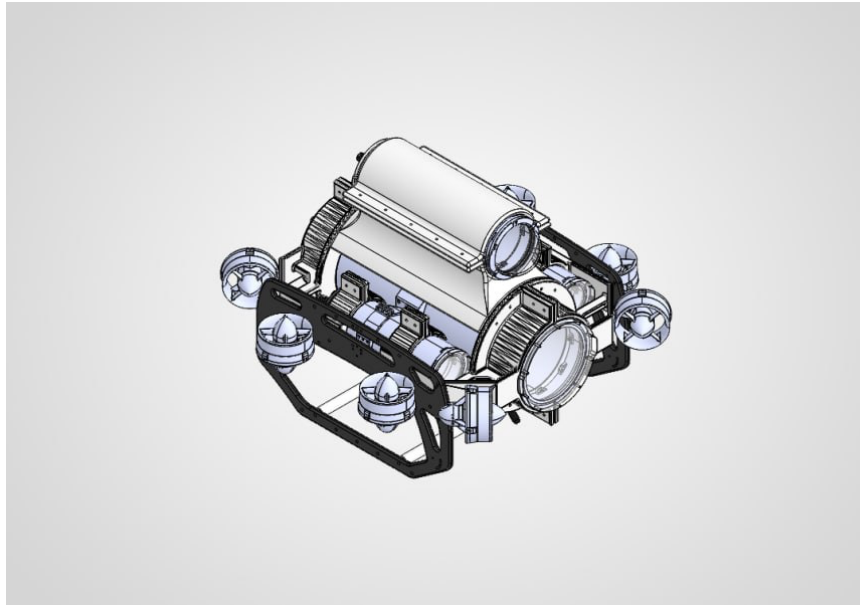


Figure 18. Riptide CAD model.